



Risk optimizations on basis portfolios: The role of sorting[☆]

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ARTICLE INFO

Keywords:

Bootstrap
Mean–variance efficiency
Portfolio sorting
Risk-based optimization
Smart beta
Style investing

ABSTRACT

This paper investigates the mean–variance and diversification properties of risk-based strategies executed on style or basis portfolios. We show that the performance of these risk strategies is highly sensitive to the sorting procedure used to form the basis assets. Whereas the extant literature provides mixed support for the outperformance of smart beta strategies based on scientific diversification, our designed strategies outperform both the market model and multifactor model. Our testing framework is based on bootstrapped mean–variance spanning tests and shows valid conclusions when controlling for multiple testing, transaction costs, and luck from random basis portfolio construction rules. Economically, our results are supported by diversification-based properties.

1. Introduction

Passive investors have, for more than 60 years, considered capitalization-weighted (CW) indexes as a proxy for the tangency portfolio, namely the Maximum Sharpe Ratio (MSR) portfolio. Although CW solutions provide a simple, cost-effective and intuitive manner in which to allocate stocks, they are exposed to certain inherent weaknesses, notably, their embedded momentum bias and their concentration in large capitalization stocks. This situation has also led to the emergence of so-called “low-risk portfolios” – such as minimum variance (MV), maximum diversification (MD), and risk parity (RP) portfolios – within the smart beta industry, as the errors in estimating expected returns have been shown to pose a serious threat to the efficiency of the MVO output (Best and Grauer, 1991a,b). The extant literature has presented mixed evidence regarding the performance of these portfolios.

This paper builds long-only smart beta solutions with attractive risk–return properties and shows that the performance of low-risk portfolios is highly sensitive to the underlying assets or building blocks. We rely on the latest advances in the portfolio sorting literature to construct the basis portfolios that will make the opportunity set (Chan et al., 2009; Chen and De Bondt, 2004; Hou et al., 2018; Kogan and Tian, 2015; Lambert et al., 2020). We compare the standard way to allocate stocks into these basis portfolios (using an independent scale with NYSE breakpoints) to the more recent dependent technique, which works in successive subportfolios. Whole-sample breakpoints are jointly used with dependent sorting to obtain our opportunity set (Lambert and Hübner, 2013; Lambert et al., 2020). Hereafter, we refer to these two sets of basis portfolios as the *dependent* and *independent* basis portfolios. Using dependent sorting, the mean–variance properties of the low-risk strategies are improved, irrespective of the optimization method (the smart beta strategy) or the number of basis portfolios considered.

[☆] All remaining errors are ours.

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A standard GRS test supports the outperformance of the dependent opportunity set for constructing smart beta optimizations. However, given the limitations to unlimited lending and borrowing, investors will be interested in verifying whether the two efficient frontiers are identical (Kan and Zhou, 2012, p. 147). Our empirical approach therefore relies on the mean–variance spanning test of Kan and Zhou (2012), which evaluates the benefit of adding new investments to a baseline portfolio and discriminates these new investments according to their contribution to the baseline portfolio's mean–variance efficiency, testing the conditions on the GMV and tangency portfolios separately for each strategy. We perform the mean–variance spanning test in a bootstrap setting, similar to that of Fama and French (2010) and Harvey and Liu (2021), to obtain spanning tests robust to the effect of multiple testing.

Our results have implications for future research on constructing factor models. Grinblatt and Saxena (2018) established a statistical technique to create a mean–variance efficient (MVE) portfolio starting from a set of characteristics or style portfolios. This portfolio is shown to span the opportunity set formed from a 3-factor model (Fama and French, 1993). Our results suggest that the GS optimization could be improved with basis portfolios formed using dependent sorting.

Our empirical study proceeds as follows. First, we build on the “diversification return” from Booth and Fama (1992) and the extensions of Willenbrock (2011) and Erb and Harvey (2006) to infer the diversification properties of our basis portfolios. We show that risk optimizations on dependent-basis portfolios outperform (in terms of the Sharpe ratio and alpha) risk optimizations on independent-basis portfolios. Second, we show that risk optimizations on dependent-basis portfolios span the investment universe of the traditional Fama and French (1993) three-factor model. Third, we apply the factor selection technique of Harvey and Liu (2021) to conduct a horse race between the different configurations of sorting methods as well as the smart beta strategies.

Our findings show that the MVE of the strategy is first subject to the definition of the sorting method and then to the choice of the risk optimization method. In particular, the MD strategy on the $3 \times 3 \times 3$ portfolio as well as the MV portfolio (especially on 2×3 and 3×3) deliver very strong alphas and diversification returns over the different periods. Our results are robust to the inclusion of transaction costs (see, e.g., Hasbrouck, 2009; Novy-Marx and Velikov, 2016). Finally, we demonstrate that our results do not hold for random sorting into portfolios, which suggests that efficient sorting procedures are important when characteristics are significantly correlated with returns.

Our approach is original and important, as it relies on scientific diversification methods but is also driven by economic insights and market practice. Indeed, the inflation of multifactor models and risk factors have extended the investor's opportunity set to style portfolios. Dimson et al. (2017) record over 6000 ETFs/ETPs with 145 smart beta equity providers across 32 different countries in 2016.⁴ Our research provides the necessary complement to extant research, which is mainly focused on low-risk strategies for individual assets, by examining the performance of smart beta solutions when applied to basis portfolios.

Finally, even though our approach has a mean–variance focus, we check the higher-moment properties of our candidates and show that they manage to reduce left-asymmetry.⁵

The rest of the paper is organized as follows. Section 2 describes the data and methodology used to construct the basis portfolios. Section 3 presents smart beta strategies and their diversification properties. In Section 4, mean–variance spanning tests are used to compare smart strategies against single-index and multifactor models. In Section 5, we test the significance of our smart strategies in complementing a multifactor model and explain the cross-section of characteristic-sorted portfolios. Section 6 investigates the performance of smart betas on portfolios constructed randomly. Section 7 concludes the paper.

2. Investment opportunity set

This section describes our opportunity set, i.e., the set of portfolios that constitute our basis assets. Our approach consists of stratifying the U.S. stocks universe in investment style portfolios under a classic angle, namely, size, book-to-market, and momentum portfolios.

Grouping stocks into portfolios offers several advantages. First, forming groups of stocks into style portfolios circumvents the burden of estimating a large covariance matrix of returns (Berk, 2000; Ao et al., 2018). Moreover, our framework is consistent with the stylized facts of Barberis and Shleifer (2003), who demonstrate the natural tendency of investors to allocate funds according to asset categories, and Froot and Teo (2008), who also observe that institutional investors tend to reallocate their funds across style groupings. Our objective of performing risk optimization techniques on investment style portfolios is, therefore, in line with the reallocation practice of institutional investors and avoids the implementation costs of working with a wide variety of individual securities.

Our stratification relies on two sorting methodologies. The first construction methodology is based on independent sorting of stocks into portfolios with NYSE-breakpoints and has become a standard in the asset-pricing literature for constructing characteristic-sorted portfolios (Fama and French, 1993, 1995, 2015). The second sorting methodology follows Lambert et al. (2020) and applies a dependent sorting method that uses whole-sample breakpoints, and this strategy implies the sorting of stocks into successive subportfolios according to their characteristics. We stratify the U.S. stock universe into six (2×3), nine (3×3) or twenty-seven ($3 \times 3 \times 3$) groups. Double sorting is performed on size and book-to-market characteristics, while the $3 \times 3 \times 3$ split is constructed for the momentum, firm size, and book-to-market characteristics. More details regarding the two methodologies can be found below.

⁴ Quoted from Bloomberg, the number of ETFs reached roughly 5,000 in 2016 while outnumbering the number of listed securities, which was slightly above 4000 over that year. This article is accessible at the following address: <https://www.bloomberg.com/news/articles/2017-05-12/there-are-now-more-indexes-than-stocks>.

⁵ For instance, maximum diversification optimizations on dependent portfolios offer, on average, Sharpe ratios (SR) adjusted for skewness and kurtosis that are 1.3 greater than those on independent portfolios for the period ranging from July 1993 to December 2015. Adjusted Sharpe ratios are estimated by $SR(1 + (S/6)SR - (K - 3/24)SR^2)$, where S and K denote skewness and kurtosis, respectively. For the sake of brevity, we do not report these descriptive statistics. However, the results are available upon request.

2.1. Data

The data are obtained by merging data from the Center for Research in Security Prices (CRSP) and Compustat. The CRSP database contains historical price information, whereas Compustat provides accounting information for all stocks listed on the major U.S. stock exchanges. The sample period ranges from July 1963 to December 2015 and covers all stocks listed on the NYSE, AMEX, and NASDAQ.⁶ For stocks listed on the NASDAQ, the data collection starts in 1973. The analysis covers a total of 630 monthly observations. Following Fama and French (1993) to filter the database and construct cross-sectional portfolios, we keep the stocks with a CRSP share⁷ code (SHRCD) of 10 or 11 at the beginning of month t , an exchange code (EXCHCD) of 1, 2 or 3 available shares (SHROUT) and price (PRC) data at the beginning of month t , available return (RET) data for month t , at least 2 years of listing on Compustat to avoid survival bias and a positive book-equity value at the end of December of year $y - 1$. We define the book value of equity as the Compustat book value of stockholders' equity (SEQ) plus the balance-sheet deferred taxes and investment tax credit (TXDITC). If available, we decrease this amount by the book value of the preferred stock (PSTK). If the book value of stockholders' equity (SEQ) plus the balance-sheet deferred taxes and investment tax credit (TXDITC) is not available, we use the firm's total assets (AT) minus its total liabilities (LT).

Book-to-market equity (B/M) is the ratio of the book value of equity for the fiscal year ending in the calendar year $y - 1$ to the market equity. Market equity is defined as the price (PRC) of the stock times the number of shares outstanding (SHROUT) at the end of June y to construct the size characteristic and at the end of December of year $y - 1$ to construct the B/M ratio. Momentum is defined as in Carhart (1997), i.e., based on a $t - 2$ until $t - 12$ cumulative prior return.

2.2. Sorting out stocks

In the original Fama–French approach, portfolios are constructed using a 2×3 independent sorting procedure, i.e., two-way sorting (small and large) on market capitalization and three-way sorting (low, medium, high) on the book-to-market equity ratio. Six portfolios are constructed at the intersection of the 2×3 classifications and are rebalanced on a yearly basis at the end of June. These style classifications are defined according to the NYSE stock exchange only and then applied to the whole sample (AMEX, NASDAQ, and NYSE).⁸ The authors chose to use NYSE breakpoints to have approximately the same market capitalization across portfolios and the same number of NYSE firms in each portfolio.

The second sorting methodology is an extension of the Fama–French sorting methodology. Lambert et al. (2020) sort stocks into successive subportfolios according to various characteristics; moreover, they define the sorting breakpoints based on the whole sample rather than considering only the NYSE. The authors indeed uncover that these NYSE breakpoints create an imbalance in the (total) number of stocks between small- and large-cap portfolios such that independent sorting leads to a higher number of stocks in small-value portfolios (Cremers et al., 2012). Since from January 1963 to December 2015, the market equity and book-to-market equity of a firm were, on average, negatively correlated (−5%), using independent sorting on negatively correlated variables can induce, by design, a strong tilt toward the extreme categories of inverse ranks, i.e., low-high and high-low.

Another practical consequence when sorting stocks into portfolios, as already stated by Chan et al. (2009), is that the original independent sorting with NYSE-breakpoints procedure could cause large-value stocks to be categorized as growth stocks. Supportive evidence can be found in the recent work of Lettau et al. (2018), who characterize the holdings of value mutual funds using Daniel et al. (1997)'s methodology.⁹ Lettau et al. (2018) show that value mutual funds tend to hold a large proportion of their investments in growth stocks. However, the ranking into quantiles relies on NYSE breakpoints. Lambert et al. (2020) document that the choices underlying the sorting methodology are important for drawing robust inferences regarding firm style characteristics. In particular, the standard procedure of NYSE breakpoints and the sequence of the dependent sort matter. If the sorting methodology is responsible for these empirical results, we claim that forming basis portfolios using this procedure will lead to a biased allocation of stocks into style portfolios, stratification of the U.S. equity universe and, therefore, to a misleading optimization exercise.

To better understand the problem, we compare the Morningstar style classification of 8,739 mutual funds (focused on the U.S. equity market) to that implied by the dependent on all breakpoints and independent on NYSE breakpoints sorting procedures. For the dependent sort, the classification of stocks by growth and value characteristics is achieved by applying a first sort based on the size characteristic of a firm and then performing a second sort based on the book-to-equity market of a firm. We construct a matrix of 5×5 portfolios along the size and value characteristics of a firm. For the independent sort, the output is similar to the 5×5 size and value portfolios available on Kenneth French's website.

The sample of mutual funds is obtained from the merger of Morningstar and CRSP Mutual Fund databases over the period from April 2002 to December 2015. The databases are merged according to the following two labels: the fund's CUSIP and a phrase matching technique applied to the fund name. Monthly performances and quarterly holdings are obtained from the CRSP Mutual

⁶ Compustat and CRSP data are available from January 1950 and January 1926, respectively. After correcting the databases for survival and backfill biases, the sample starts in July 1953. For comparison purposes, we start our empirical analyses from July 1963 onwards, as in Fama and French (1993).

⁷ See Hasbrouck (2009, p. 1455): "restricted to ordinary common shares (CRSP share code 10 or 11) that had a valid price for the last trading day of the year and had no changes of listing venue or large splits within the last 3 months of the year".

⁸ The NYSE is represented by stocks that account for the largest capitalization in the CRSP database. The exchange codes 1, 2 and 3 represent the NYSE, NASDAQ, and AMEX, respectively.

⁹ Daniel et al. (1997) sort stocks at the end of June of each year to form 125 portfolios along a triple dependent sort, with the first sort based on the firm's size, the second sort based on the firm's industry-adjusted book-to-market ratio and the final sort based on the firm's momentum (cumulative return from $t-2$ to $t-12$).

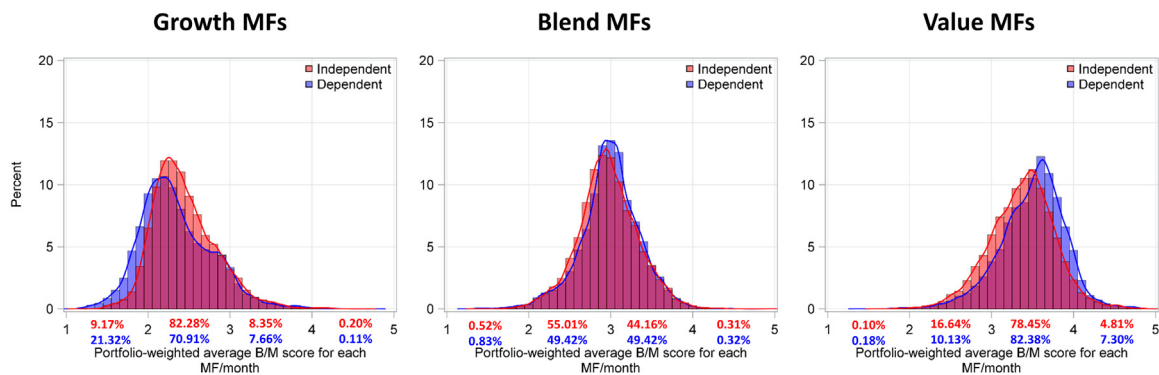


Fig. 1. Distribution of BM-scores of Mutual Funds: Independent vs. Dependent Sorts. The figure shows the kernel distribution in the BM-scores of mutual funds with a focus on the U.S. equity market for which Morningstar attributes a value-growth classification. The value-growth classification is applied to the mutual funds present in the CRSP mutual funds database. For each point in time when a fund reports its holdings, we associate a BM-score from a 1–5 scale according to the Fama–French’s 5×5 size and value *independent* sorting methodology or a 5×5 size and value *dependent* sorting methodology. The fund’s BM-score is then calculated as the percentage of the total net assets (TNA) weighted average of the 1–5 scale of the securities the fund holds. The distributions are displayed for the following 3 Morningstar classifications of funds: growth (left), blend (middle), and value (right). The sample period ranges from April 2002 to December 2015.

Fund Database. Style classifications are obtained from Morningstar. Next, we match the information on the funds’ holdings with the value-growth classification from the independent (with NYSE breakpoints) and dependent (with whole-sample breakpoints) sorting methodologies. The classification is applied according to accounting information obtained from Compustat at the end of June for each stock. The stock universe is then split according to a 1–5 scale, as follows: 1 represents a growth tilt, 3 represents a blend/neutral style, and 5 represents a value tilt, as in the work of Lettau et al. (2018).

Fig. 1 illustrates the distribution of funds along the dependent-name breakpoints (hereafter referred to as dependent sort) and independent-NYSE (hereafter referred to as independent sort) frameworks for the following Morningstar categories: growth (left), blend (middle), and value (right). The distributions across the growth and value styles demonstrate that the BM-scores of mutual funds computed using a dependent scale instead of an independent scale are better aligned with the style classification of the fund. Indeed, the distribution of growth funds is more skewed to the left for the dependent sort, as shown by 21.32% (dependent sort) vs. 9.17% (independent sort) of the observations falling under the first quintile of the distribution. Similarly, for value funds, the distribution is more skewed to the right for the dependent sort given that 7.30% (dependent sort) vs. 4.81% (independent sort) of the observations fall under the last quintile of the distribution. Finally, the mode of the distribution of blended mutual funds under a dependent scale falls around the third quintile, as 49.42% of the observations are found in quintiles 2 and 4. Using an independent scale, the mode is shifted to values below 3, which are representative of growth stocks. This would wrongly indicate that these funds hold more growth than value stocks and suggest that the sorting definition could mislead the allocation.

In summary, value (growth) mutual funds have a higher probability of being categorized as value (growth) funds under a dependent sorting procedure than under an independent sorting procedure. Blended mutual funds also show better neutrality to the value-growth categorization using a dependent sort.

2.3. Pair-wise correlation of style portfolios

Fig. 2 illustrates the stock distribution when the number of portfolios is increased either by a larger split of the sample (from a 2×3 to a 3×3 split) or by adding a new characteristic ($3 \times 3 \times 3$). The $3 \times 3 \times 3$ splits are constructed based on the size, value, and momentum characteristics of a firm. We observe that using an independent sort results in an imbalance of stocks across the portfolios, and this effect becomes larger when more groups are constructed.

We expect the higher level of diversification induced by the dependent sort and by the higher-dimensional space representation of the U.S. equity market to deliver additional diversification benefits for risk-based optimizations with regard to independent-basis portfolios.

To verify this hypothesis, in Table 1, we compute the average correlation between the investment style portfolios. It can be observed that the correlation is lower when stocks are sorted dependently and are split into a larger number of groups (i.e., $3 \times 3 \times 3$). Here, the basis portfolios are cap-weighted portfolios to mitigate the impact of small-cap stocks and are rebalanced annually at the beginning of July, which is consistent with the approach of Fama and French (1993).¹⁰

¹⁰ Our work could be further extended as in the work of Brandt et al. (2009), who allocated the weights of stocks in portfolios according to the level of their characteristics to maximize a CRRA investor’s utility. However, we leave this option to more interested readers, as we are more interested in reviewing the consequences of stock classification methods for risk-based optimizations rather than the allocation scheme inside the basis portfolios.

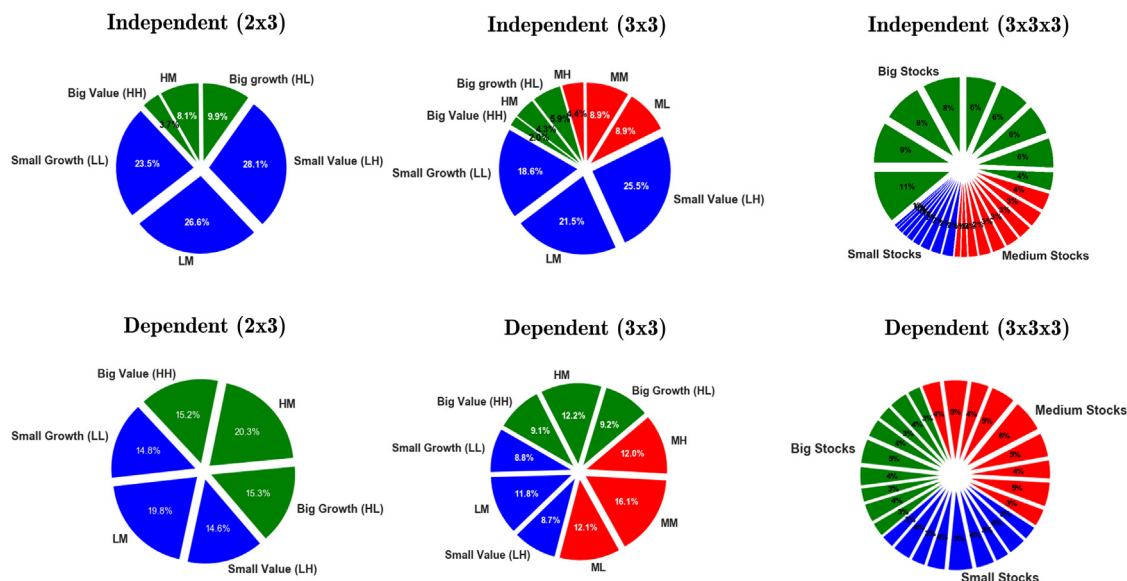


Fig. 2. Average Stock Distribution with Independent vs Dependent Sorting. These plots show the stock distribution among the 2×3 and 3×3 characteristic-sorted portfolios based on size (low, medium and high) and the book-to-market equity ratio (low, medium and high) for the independent- and dependent-sorting methodologies. We also report the average percentage of stock distributions among the $3 \times 3 \times 3$ characteristic-sorted portfolios when momentum is added as a third variable. For clarity, we group the 27 portfolios according to their size classifications (small, medium, and large). The period is the interval from July 1963 to December 2015.

Table 1
Correlation between characteristic-sorted portfolios.

#Number of portfolios	Independent Sort (1)	Dependent Sort (2)	Difference (1)–(2)
Panel A: Average correlation (in %)			
2×3	83.85	77.24	6.61
3×3	83.19	73.79	9.40
$3 \times 3 \times 3$	73.84	62.78	11.06
Panel B: Standard deviation of correlation (in %)			
2×3	7.15	11.11	–3.96
3×3	8.17	13.84	–5.67
$3 \times 3 \times 3$	10.67	17.10	–6.44

The table reports the average correlation (Panel A) and the standard deviation of the correlations (Panel B) for the characteristic-sorted portfolios constructed using independent and dependent sorting methodologies. The third column specifies the difference between the independent and dependent sorting results. Correlations are estimated based on rolling windows of 60 days, as expressed in percentages, and the sample period extends from 01/07/1963 to 31/12/2015.

3. Smart investment strategies

Table 2 displays the analytic forms of the risk-based allocations that serve as a practical base in our empirical analysis, namely, *minimum variance* (MV), *maximum diversification* (MD), and *risk parity* (RP). Following Ao et al. (2018), Ardia et al. (2018), Grinblatt and Saxena (2018), Roncalli and Weisang (2016), among others, these risk-based allocations are rebalanced on a monthly basis.

To feed these low-risk investment strategies (MV, MD, RP), we form 6, 9, and 27 cap-weighted portfolios and use 60 daily returns to estimate the covariance matrix.¹¹ In the most extreme case (27 portfolios), we are left with 0.17 data points per parameter. Even this simplified situation might create large sampling errors if we consider only the sample covariance matrix in our optimizations. In our applications, we use a traditional shrinkage methodology developed by Ledoit and Wolf (2004) to estimate the covariance matrix with lower sampling errors. Further details on the shrinkage method used can be found in Appendix A.

¹¹ We use a 60-day range to estimate the variance–covariance matrices for two reasons. First, Fama and French (2018) use 60 days of lagged returns to estimate the monthly variance of stocks, and second, real-life applications regarding tradable assets would also impose practical constraints over the length available for the time-series (Idzorek and Kowara, 2013).

Table 2

List of the smart beta strategies' objective functions.

Strategy	Referenced authors	Objective Function
Minimum Variance (MV)	Clarke et al. (2013)	$\min f(w) = \sum_i^N \sum_j^N w_i \sigma_{ij} w_j$
Maximum Diversification (MD)	Choueifaty and Coignard (2008)	$\max f(w) = \frac{\sum_i^N w_i \sigma_i}{\sqrt{\sum_i^N \sum_j^N w_i \sigma_{ij} w_j}}$
Risk parity (RP)	Maillard et al. (2010)	$\min f(w) = \sum_i^N \sum_j^N (w_i \times (\Sigma w)_i - w_j \times (\Sigma w)_j)^2$

The table decomposes the smart beta strategies' objective function applied to the constituents' weights. The first column refers to the common name of the strategy. The second column specifies the main authors who have analyzed the strategy. The third column reports the objective function for minimization or maximization. In the objective function, w refers to the weights, N is the total amount of assets introduced in the optimization, i and j denote the i -th asset and the j -th asset, respectively, σ_{ij} is the covariance between the i -th asset and j -th asset, p refers to the portfolio, and $(\Sigma w)_i$ is the risk contribution of the i -th asset. All the objective functions are subjected to long-only budget constraints, i.e., $w_i \in [0.01, 0.99]$ and $\sum_{i=1}^N w_i = 1$.

Table 3

Summary statistics of smart beta portfolios.

	Performance		Basis portfolio weights			
	Mean (in %)	STD (in %)	$\sigma(w)$ (in %)	$w = 0.01$ (in %)	$ w - \mu(w) > 1.96 \times \sigma(w)$ (in %)	1/HI
Panel A: Independent						
MD _{2 × 3}	1.090	4.698	5	2	15	5.6
MD _{3 × 3}	1.114	4.772	6	12	29	7.5
MD _{3 × 3 × 3}	1.090	4.855	4	61	58	16.3
MV _{2 × 3}	1.167	4.577	25	14	45	2.3
MV _{3 × 3}	1.153	4.743	17	10	52	3.2
MV _{3 × 3 × 3}	1.147	4.532	7	2	91	6.8
RP _{2 × 3}	1.104	4.720	3	0	19	5.8
RP _{3 × 3}	1.122	4.845	2	0	28	8.6
RP _{3 × 3 × 3}	1.136	4.868	1	0	65	25.4
Long-only MVE						
MVE _{2 × 3}	1.322	5.054	32	42	70	1.4
MVE _{3 × 3}	1.367	5.211	26	78	83	1.6
MVE _{3 × 3 × 3}	1.323	5.241	12	100	98	2.9
Panel B: Dependent						
MD _{2 × 3}	1.181	4.878	10	3	33	4.7
MD _{3 × 3}	1.190	5.010	10	44	48	5.5
MD _{3 × 3 × 3}	1.223	5.272	5	87	80	10.9
MV _{2 × 3}	1.310	5.075	26	15	50	2.2
MV _{3 × 3}	1.263	5.061	17	12	50	3.4
MV _{3 × 3 × 3}	1.179	4.905	7	3	88	7.6
RP _{2 × 3}	1.148	4.914	5	0	28	5.6
RP _{3 × 3}	1.168	5.136	3	0	33	8.3
RP _{3 × 3 × 3}	1.194	5.140	1	0	60	24.0
Long-only MVE						
MVE _{2 × 3}	1.634	5.344	33	55	78	1.3
MVE _{3 × 3}	1.674	5.444	25	77	76	1.7
MVE _{3 × 3 × 3}	1.503	5.672	11	100	93	3.6

The table reports the summary statistics for the smart beta strategies (MD, MV, RP and the long-only MVE) with regard to their risk and return (mean, standard deviation), as well as regarding their weighting scheme on the basis portfolios. $\sigma(w)$ stands for the volatility of the weighting scheme among the basis portfolios for each strategy. $w = 0.01$ is the frequency at which a smart beta strategy puts a minimal weight of 1% on one of each type of basis portfolio. $|w - \mu(w)| > 1.96 \times \sigma(w)$ is the frequency at which a smart beta strategy significantly rebalances its weighting scheme across the basis portfolios, i.e., the weight is superior to a confidence interval centered around the average weight. Finally, the last column presents the average effective number of basis portfolios of each strategy over the period (measured by the inverse of the Herfindahl index (HI)). Panel A (resp. B) considers an independent (resp. dependent) opportunity set. These strategies are rebalanced on a monthly basis, and the number of portfolios is six (2×3), nine (3×3) or twenty-seven ($3 \times 3 \times 3$). All estimates are obtained from monthly values. The sample period extends from July 1963 to December 2015.

3.1. Summary statistics

Table 3 presents the summary statistics for our smart beta portfolios constructed using an independent or dependent opportunity set. We also report the results for a mean–variance (MVE) strategy for which the expected returns are estimated from the historical sample mean and the weights are constrained to include long-only.

The smart beta portfolios are compared first based on their mean (gross returns) and variance. The smart beta strategies presents higher returns and higher volatility under the dependent framework than under the independent framework (see Table 4 for tests of differences of Sharpe ratios under the two frameworks).

Table 4
Statistics of smart beta portfolios: Net of transaction costs.

	Transaction costs (in %)			Turnover (in %)			Sharpe (Gross return)			Sharpe (Net return)		
	Dep	Ind	Δ Dep-Ind	Dep	Ind	Δ Dep-Ind	Dep	Ind	Δ Dep-Ind	Dep	Ind	Δ Dep-Ind
MD _{2 × 3}	0.009	0.003	0.006***	23.14	13.58	9.56***	0.159	0.147	0.013***	0.158	0.146	0.012***
MD _{3 × 3}	0.014	0.005	0.009***	31.60	21.99	9.61***	0.157	0.149	0.008***	0.154	0.148	0.006***
MD _{3 × 3 × 3}	0.019	0.009	0.01***	40.73	40.60	0.13	0.155	0.142	0.013***	0.152	0.140	0.012***
MV _{2 × 3}	0.019	0.011	0.008***	48.37	47.75	0.62	0.179	0.167	0.012***	0.175	0.165	0.010***
MV _{3 × 3}	0.030	0.018	0.013***	66.28	69.52	−3.24*	0.170	0.159	0.011***	0.164	0.155	0.009***
MV _{3 × 3 × 3}	0.030	0.019	0.012***	74.85	77.10	−2.25*	0.158	0.164	−0.006**	0.152	0.160	−0.008***
RP _{2 × 3}	0.005	0.002	0.003***	13.79	9.83	3.96***	0.152	0.149	0.003***	0.151	0.148	0.002***
RP _{3 × 3}	0.006	0.003	0.004***	14.14	10.18	3.96***	0.149	0.149	0.0002	0.148	0.148	−0.0005
RP _{3 × 3 × 3}	0.007	0.003	0.004***	16.09	12.52	3.56***	0.154	0.151	0.003***	0.153	0.150	0.003***
Long-only MVE												
MVE _{2 × 3}	0.040	0.032	0.008***	56.92	78.45	−21.53***	0.230	0.182	0.048***	0.222	0.176	0.046***
MVE _{3 × 3}	0.058	0.036	0.021***	77.82	78.73	−0.92**	0.233	0.185	0.048***	0.223	0.179	0.044***
MVE _{3 × 3 × 3}	0.066	0.039	0.027***	79.01	87.31	−8.29***	0.194	0.176	0.018***	0.182	0.168	0.014***

The table presents the summary statistics for the transaction costs, turnover, and (gross and net) Sharpe ratios of smart beta portfolios (MD, MV, and RP, as well as the long-only maximum Sharpe ratio (MVE)) under a dependent (Dep) and an independent (Ind) framework. These strategies are rebalanced on a monthly basis, and the number of portfolios is six (2 × 3), nine (3 × 3) or twenty-seven (3 × 3 × 3). For each strategy, the spread between the two frameworks is reported, as well as the *p*-value of the difference test. The difference tests for the transactions costs and turnovers are obtained from a sample *t*-test. The difference test on the Sharpe ratio used the approach of [Ledoit and Wolf \(2008\)](#), for which the MATLAB code is available at Prof. Wolf's [website](#). All estimates are obtained from monthly values. The sample period extends from July 1963 to December 2015.

*Denote significance levels of 10%.

**Denote significance levels of 5%.

***Denote significance levels of 1%.

The average weights of these strategies over the long period correspond to 1/*N* under both frameworks. We therefore provide the distribution of the weights of smart beta strategies among basis portfolios, especially the following:

- The volatility of the weighting scheme among the basis portfolios for each strategy.
- The frequency at which a smart beta strategy puts a minimal weight of 1% ($w = 0.01$) on one of each type of basis portfolio.
- The frequency at which a smart beta strategy significantly rebalances its weighting scheme across the basis portfolios; i.e., the weight is superior to a confidence interval centered on the average weight ($|w - \mu(w)| > 1.96 \times \sigma(w)$).
- The average effective number of basis portfolios of each strategy over the period (as measured by the inverse of the Herfindahl index). The indicator is given by the following:

$$\frac{1}{T} \sum_t \frac{1}{\sum_i^N w_i^2}, \quad (1)$$

where *N* is the number of basis portfolios and *T* is the number of monthly observations.

The interpretation of the average effective number of basis portfolios is as follows: the MD_{2×3} strategy developed under an independent framework invests on average in approximately 5.6 basis portfolios out of the 6 constituent portfolios. Its equivalent under the dependent framework holds 4.7 basis portfolios. With regard to their independent counterparts, lower diversification can indeed be found for the dependent MD and RP strategies, while slightly higher diversification can be found for the dependent MV and MVE portfolios.

These different criteria used to capture the distribution of weights show that the dependent framework provides slightly more dynamic allocations (see the volatility of weights or the percentage of significant rebalancing). Additionally, [Table 1](#) shows that the average correlations across basis portfolios are lower but more time-varying under a dependent framework than under an independent framework. This also means that the dependent framework rebalances more to take advantage of this changing correlation structure among its basis portfolios. The levels of diversification of the dependent and independent frameworks are quite similar but differ across smart beta strategies.

[Table 4](#) provides comparative information on the turnover within the basis portfolio constituents between the dependent and independent framework, as well as comparative measures of transactions costs and Sharpe ratios. Following [DeMiguel et al. \(2009\)](#), the turnover is given by

$$\frac{1}{T} \sum_1^T \sum_{j=1}^K |w_j^{t+1} - w_j^t|, \quad (2)$$

where w_j^{t+1} and w_j^t are the weights attributed to the stock portfolio j at the beginning of the period $t + 1$ and just *before* rebalancing, respectively. The transaction costs are obtained based on the [Hasbrouck \(2009\)](#) estimates; technical details can be found in [Appendix D](#). Difference tests are performed on the turnover¹² and the Sharpe ratio estimates under both frameworks.¹³

[Table 4](#) shows that the dependent framework induces higher transaction costs mainly due to more frequent rebalancing of stocks within the basis portfolios. The spreads range from 0.3 to 1.3 basis points per month for the pure smart beta strategies (MD, MV, and RP). The dependent framework produces basis portfolios that have significantly higher turnover values. This can be explained by the use of whole-sample breakpoints that will adjust the composition of the portfolios according to economic conditions.

This difference in transaction costs might affect the superiority of the dependent framework. The table performs difference tests on the Sharpe ratios defined for both the net and gross transaction fees. Both stay significantly higher under the dependent framework.

3.2. Diversification properties

This section compares the diversification returns achieved by implementing the risk-based optimization approach based on dependent- and independent-basis portfolios, further decomposes the diversification return into its two components and performs a paired difference test.

The diversification return, according to [Booth and Fama \(1992\)](#), is defined as the difference between the compound return of a portfolio and the weighted average of the compound return of its constituent assets. This relationship assumes that the portfolios are rebalanced so that the weights are held constant and moments higher than the second are very small. In this situation, the diversification return increased with the spread between the individual asset variance and its covariance with the portfolio.

Denoting the geometric average return as g , the volatility as σ , and the arithmetic average return as μ , the geometric return of a portfolio p can be expressed as follows:

$$g_p = \mu_p - \frac{\sigma_p^2}{2}. \quad (3)$$

The diversification return (DR) can be written as follows ([Booth and Fama, 1992](#); [Willenbrock, 2011](#)):

$$DR = g_p - \sum_i^N w_i g_i, \quad (4)$$

where i stands for the i -th security in the portfolio p , and g refers to the geometric return. The weights (w_i) are assumed to be constant over the estimation period. We refer to the fixed-weight diversification return using the superscript (FW).

Substituting (3) in (4), we obtain the following

$$DR^{FW} = \mu_p - \frac{\sigma_p^2}{2} - \sum_i^N w_i \left(\mu_i - \frac{\sigma_i^2}{2} \right). \quad (5)$$

Rearranging the terms,

$$DR^{FW} = \underbrace{\mu_p - \sum_i^N w_i \mu_i}_{DR_1^{FW} = 0 \text{ if weights are constant}} + \underbrace{\frac{1}{2} \left(\sum_i^N w_i \sigma_i^2 - \sigma_p^2 \right)}_{DR_2^{FW} = \text{variance reduction benefit}}. \quad (6)$$

In the last part of the equation, we retrieve the variance reduction benefit (DR_2^{FW}) of [Booth and Fama \(1992\)](#) and [Willenbrock \(2011\)](#). Note that in theory, w_i should be determined at inception and remain constant over the life of the strategy. To implement equation (6) for rebalancing strategies (non-fixed weight), [Erb and Harvey \(2006\)](#) use the average of the weights over the sample period ($\bar{w}_i = \frac{1}{T} \sum_1^T w_i^t$). As the computation of the diversification return induces a comparison with a static portfolio endogenous to each strategy, it is difficult to compare the total diversification gains across a pair of smart beta strategies that systematically shift the assets weights. Therefore, we extend equation (6) to consider a rebalanced portfolio p and its diversification return with regard to an EW benchmark. We chose the equal-weighted strategy because this is the only allocation for which we know ex-ante the value of w_i , that is $(1/N)$, as long as the amount of securities (N) remains constant in the portfolio.¹⁴ In this alternative framework, we impose that two smart beta strategies constructed on an equivalent number of basis portfolios (N) share the same benchmark $(1/N)$. We denote the principle that the diversification return is compared to an EW strategy using the superscript (EW), as follows:

$$DR^{EW} = \underbrace{\mu_p - \frac{1}{N} \sum_i^N \mu_i}_{DR_1^{EW}} + \underbrace{\frac{1}{2} \left(\frac{1}{N} \sum_i^N \sigma_i^2 - \sigma_p^2 \right)}_{DR_2^{EW}}. \quad (7)$$

¹² The difference test is performed with a sample t -test.

¹³ We follow the [Ledoit and Wolf \(2008\)](#) bootstrap test for comparing the *ex post* Sharpe ratios of smart beta strategies when performed on distinct sets of basis portfolios, i.e., dependent vs. independent. The number of bootstraps is set to 5,000 with blocks of 10 observations, as recommended by the authors. The method is explained in [Appendix B](#).

¹⁴ Due to the simplicity and the out-of-sample performance of the strategy, [DeMiguel et al. \(2009, p. 1948\)](#) also recommend the “ $1/N$ ” portfolio as “the first obvious benchmark” for evaluating other weighting schemes.

With this benchmark, it may not be immediately clear whether the measure departs from the essence of diversification, which concerns the interaction among the constituents of a single portfolio. We may see this interaction as DR^{EW} , which simply adds another strategic return, i.e., $\sum_i^N \left(\bar{w}_i - \frac{1}{N}\right) g_i$, to DR^{FW} , that is,

$$\begin{aligned}
 DR^{EW} &= DR^{FW} + \underbrace{\sum_i^N \left(\bar{w}_i - \frac{1}{N}\right) g_i}_{DR_3^{FW} = \text{gain in strategic return over EW}} \\
 &= DR^{FW} + \sum_i^N \left(\bar{w}_i - \frac{1}{N}\right) \left(\mu_i - \frac{\sigma_i^2}{2}\right) \\
 &= DR^{FW} + \sum_i^N \left(\bar{w}_i - \frac{1}{N}\right) \mu_i - \frac{1}{2} \sum_i^N \left(\bar{w}_i - \frac{1}{N}\right) \sigma_i^2 \\
 &= DR^{FW} + \sum_i^N \bar{w}_i \mu_i - \frac{1}{N} \sum_i^N \mu_i - \frac{1}{2} \sum_i^N \bar{w}_i \sigma_i^2 - \frac{1}{2} \frac{1}{N} \sum_i^N \sigma_i^2 \\
 &= \mu_p - \frac{1}{N} \sum_i^N \mu_i + \frac{1}{2} \left(\frac{1}{N} \sum_i^N \sigma_i^2 - \sigma_p^2 \right) = DR_1^{EW} + DR_2^{EW},
 \end{aligned} \tag{8}$$

where the term DR_3^{FW} is the spread between the weighted average and the simple average of the geometric return of the portfolio's assets. It can be interpreted as the hypothetical gain in geometric return obtained by selecting a rebalancing strategy different from an equal-weighted allocation. The term might thus be useful to measure the return attributed to the strategic decision, which allocates the opportunity set. For example, it could tell us whether a *minimum variance* offers a higher strategic return than an *equal-weight* for allocating a set of independent-sorted portfolios (or dependent-sorted portfolios).

To test the statistical difference in a diversification return brought by a pair of strategies performed on two opportunity sets, we follow the indirect bootstrap framework of [Ledoit and Wolf \(2008\)](#), which was initially constructed to determine whether a pair of strategies have statistically equivalent Sharpe ratios. In their conclusion, Ledoit and Wolf suggest extending their model to other mean–variance performance measures. We thus revisit their framework for use on a spread in diversification return between a pair of strategies. We provide more details on our extension in [Appendix B](#). In short, we aim to compare the spread in diversification return ($\Delta \text{Dep-Ind}$) estimated in the original sample to an empirical distribution of spreads constructed from bootstrapped samples and then infer the level of significance of this spread.

In [Table 5](#), we report the results of the diversification return for the low-risk investment strategies based on 2×3 , 3×3 , and $3 \times 3 \times 3$ basis portfolios. We observe that 8 out of 9 times, the spread in the variance reduction benefit (DR_2^{FW} or DR_2^{EW}) is statistically greater for the set of dependent-basis portfolios at the usual significance level. According to DR_3^{FW} , it also seems more interesting to perform any smart beta strategy on dependent portfolios rather than on independent portfolios.

For 7 out of 9 risk-based optimizations, the dependent opportunity set offers significantly higher total diversification returns (DR^{EW}) than the independent sort (see Panel B). The annualized diversification return is on average 1.59% (resp. 0.70%) across smart beta strategies when using dependent (resp. independent) sorting. To compare our results to [Grinblatt and Saxena \(2018\)](#)'s framework, the last three lines of the table also provide the special case of the long-only MVE strategy. It shows that the dependent-MVE also provides significant improvement over its independent equivalent. However, the improvement is mostly driven by the return improvement (DR_1) compared to other smart beta strategies (as expected, given that MV, MD and RP do not rely on expected return estimates).

Consistent with [Grinblatt and Saxena \(2018\)](#), a risk–return improvement can be achieved by allocating basis portfolios with optimization techniques, which is a departure from the traditional equal-weight allocation. However, our results reveal that this is only valid when advanced sorting methods are used to construct the basis portfolios. In the table, the dependent opportunity sets systematically outperform the independent opportunity sets, even in the case of the long-only MVE portfolio.

The next section is dedicated to providing a methodological analysis of the mean–variance performance of smart beta strategies.

4. Mean–variance spanning test

Mean–variance spanning à la [Huberman and Kandel \(1987\)](#) means that a set of K risky assets spans a larger set of $K + N$ assets if the efficient frontier made of the K assets is identical to the efficient frontier comprising the $K + N$ assets. We initially set R_1 to a K -vector of the returns on K benchmark assets, R_2 to a N -vector of the returns on N test assets, and R to the raw returns on $K + N$ assets. [Huberman and Kandel \(1987\)](#) define the following regression test:

$$R_2^t = \alpha + \beta R_1^t + e^t \tag{9}$$

The null hypothesis H_0 sets $\alpha = 0$ and $\delta = 1_N - \beta 1_K = 0$ and implies that mean–variance spanning as the benchmark assets dominate the test assets. Both assets have the same mean, but the K benchmarks have a lower variance than the test assets.

Table 5
Diversification returns: Equal-weight benchmark.

Panel A: Fixed-Weight (FW) Benchmark									
	DR_1^{FW}			DR_2^{FW}			DR_3^{FW}		
	Ind	Dep	Δ Dep-Ind	Ind	Dep	Δ Dep-Ind	Ind	Dep	Δ Dep-Ind
$MD_{2 \times 3}$	−0.005	−0.003	0.002	0.025	0.038	0.012***	0.013	0.092	0.079***
$MD_{3 \times 3}$	0.004	−0.032	−0.035	0.031	0.050	0.019***	0.009	0.100	0.091***
$MD_{3 \times 3 \times 3}$	−0.041	−0.075	−0.034	0.046	0.080	0.034***	0.007	0.119	0.112**
$MV_{2 \times 3}$	0.013	−0.025	−0.038	0.024	0.031	0.007	0.053	0.213	0.162***
$MV_{3 \times 3}$	−0.001	−0.036	−0.035	0.024	0.045	0.020***	0.055	0.197	0.119***
$MV_{3 \times 3 \times 3}$	−0.018	−0.115	−0.097*	0.047	0.070	0.023***	0.04	0.154	0.089***
$RP_{2 \times 3}$	0.005	0.005	0.00	0.024	0.036	0.012***	0.078	0.240	0.035***
$RP_{3 \times 3}$	0.005	0.000	−0.005	0.027	0.042	0.014***	0.061	0.180	0.032***
$RP_{3 \times 3 \times 3}$	−0.001	−0.010	−0.009	0.043	0.065	0.023***	0.055	0.144	0.031***
Long-only MVE									
$MVE_{2 \times 3}$	0.186	0.33	0.144**	0.009	0.012	0.004	0.016	0.051	0.160***
$MVE_{3 \times 3}$	0.209	0.357	0.148**	0.01	0.025	0.015***	0.016	0.048	0.142***
$MVE_{3 \times 3 \times 3}$	0.157	0.185	0.028	0.028	0.043	0.016*	0.016	0.047	0.113***
Panel B: Equal-Weight (EW) Benchmark									
	DR_1^{EW}			DR_2^{EW}			DR_3^{EW}		
	Ind	Dep	Δ Dep-Ind	Ind	Dep	Δ Dep-Ind	Ind	Dep	Δ Dep-Ind
$MD_{2 \times 3}$	0.005	0.088	0.083**	0.027	0.038	0.011***	0.032	0.126	0.094***
$MD_{3 \times 3}$	0.010	0.070	0.060	0.033	0.048	0.015***	0.043	0.118	0.075*
$MD_{3 \times 3 \times 3}$	−0.034	0.063	0.097	0.046	0.061	0.014**	0.012	0.124	0.112*
$MV_{2 \times 3}$	0.082	0.218	0.136*	0.033	0.028	−0.004	0.115	0.246	0.131*
$MV_{3 \times 3}$	0.050	0.143	0.093	0.034	0.046	0.012*	0.084	0.189	0.105
$MV_{3 \times 3 \times 3}$	0.022	0.019	−0.003	0.061	0.080	0.018***	0.084	0.099	0.015
$RP_{2 \times 3}$	0.020	0.056	0.036**	0.026	0.036	0.010***	0.046	0.092	0.047***
$RP_{3 \times 3}$	0.019	0.048	0.029*	0.029	0.042	0.013***	0.048	0.090	0.041**
$RP_{3 \times 3 \times 3}$	0.011	0.035	0.024	0.046	0.068	0.022***	0.057	0.102	0.046***
Long-only MVE									
$MVE_{2 \times 3}$	0.238	0.541	0.304***	0.01	0.014	0.005	0.247	0.555	0.308***
$MVE_{3 \times 3}$	0.264	0.554	0.290***	0.011	0.025	0.015**	0.275	0.579	0.304***
$MVE_{3 \times 3 \times 3}$	0.198	0.343	0.145	0.027	0.039	0.012	0.225	0.382	0.157*

The table reports the return diversification from Eq. (6) (FW), Eq. (7) (EW) and Eq. (8) (EW-FW) for each risk optimization strategy (MD, MV, and RP, as well as the long-only maximum Sharpe ratio (MVE)) under a dependent (Dep) framework and an independent (Ind) framework. These strategies are rebalanced on a monthly basis, and the number of portfolios is six (2×3), nine (3×3) or twenty-seven ($3 \times 3 \times 3$). The components of the diversification return are reported as a percentage and on a monthly basis. The sample period extends from July 1963 to December 2015. For each strategy, the spread between the two frameworks is also reported, as well as the p -value of the difference test. The p -value is estimated following the approach of Ledoit and Wolf (2008), which can be found in Appendix B.

*Denote significance levels of 10%.

**Denote significance levels of 5%.

***Denote significance levels of 1%.

Considering an efficient frontier comprising $K + N$ assets, the following two formulas express the optimal weights of the N assets in the tangent MSR (Qw_1) and GMV (Qw_2) portfolios, as follows¹⁵:

$$Qw_1 = \frac{QV^{-1}\mu}{1'_{N+K}V^{-1}\mu} = \frac{\Sigma^{-1}\alpha}{1'_{N+K}V^{-1}\mu} \quad (10)$$

$$Qw_2 = \frac{QV^{-1}1_{N+K}}{1'_{N+K}V^{-1}1_{N+K}} = \frac{\Sigma^{-1}\delta}{1'_{N+K}V^{-1}1_{N+K}}$$

where $Q = [0_{N \times K}, I_N]$ with I_N , an $N \times N$ identity matrix, $\Sigma = V_{22} - V_{21}V_{11}^{-1}V_{12}$, which comes from V the variance–covariance matrix of the K benchmark assets (R_1) plus the N test assets (R_2), that is,

$$V = Var[R_1^t, R_2^t] = \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix} \quad (11)$$

The value of alpha will determine whether the tangency portfolio is improved by the introduction of the N assets, while testing beta will determine whether a significant change is induced in the GMV portfolio by the addition of the N assets. Huberman and Kandel (1987) jointly test these two conditions. The rejection of mean–variance spanning could thus have two sources: an improvement in the slope of the tangency portfolio or an improvement in the risk–return properties of the GMV portfolio. However,

¹⁵ To make a clear distinction between the risk-optimization that minimizes the portfolio variance and the ex-post global minimum variance portfolio, we denote the former as MV and the latter as GMV in the rest of the paper. MSR will denote the tangent portfolio, i.e., the maximum Sharpe ratio portfolio.

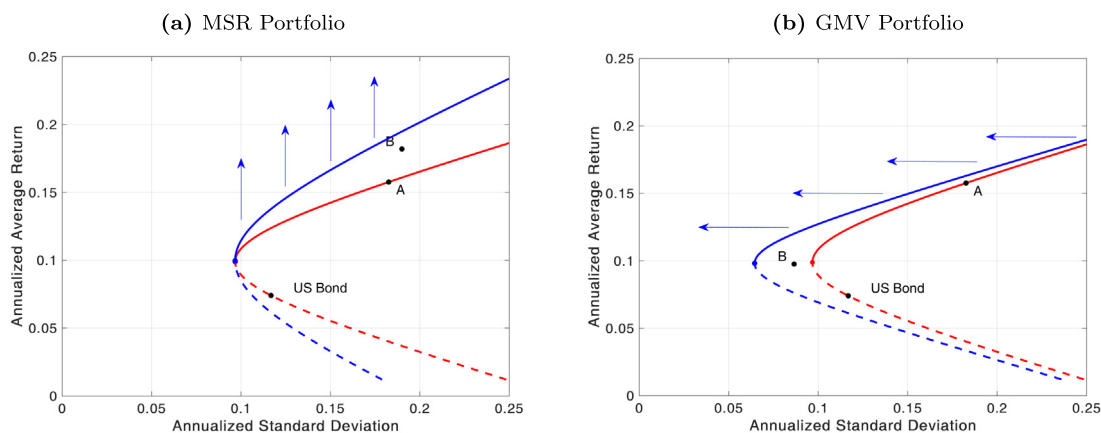


Fig. 3. Improving the MSR (a) and GMV (b) Portfolios. The figure displays the spanning illustration for the opportunity sets comprising the benchmark assets (R_1), i.e., the 30-Year U.S. Treasury Bond and Portfolio A, in the color red. The benchmark assets plus a test asset (R_2), i.e., Portfolio B, are displayed in the color blue. The x -axis reports the annualized standard deviation (in %), and the y -axis reports the annualized average return (in %). This example is fictitious but illustrates an improvement of the MSR (GMV) portfolio after Portfolio B is added to the benchmark assets in Figure A (Figure B). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

beta can be estimated more accurately than alpha, as it does not depend on the expected returns of the assets (see Eq. (10)). Therefore, statistical significance of the change in the composition of the GMV portfolio can be reached without implying economic significance. To circumvent this problem, Kan and Zhou (2012, hereafter KZ) propose testing the two conditions separately and adjusting the significance threshold of the two tests to economic significance. If the GMV condition is rejected more easily, the significance threshold should be reduced.

In Fig. 3(a), we graphically illustrate a significant improvement in the tangency portfolio when a test asset (R_2) is added to the benchmark assets (R_1). In Fig. 3(b), a significant improvement in the GMV portfolio occurs when a test asset (R_2) is added to the benchmark assets (R_1).

Our application of the mean–variance spanning test is used to determine whether smart investment strategies span the existing benchmarks, such as the single-factor model or the multi-factor model of Fama and French (1993) (Section 4.1). Spanning tests between the different configurations of low-risk portfolios are also performed to investigate the consequences of the use of different opportunity sets (Section 4.2).¹⁶

Although we assume the existence of a risk-free asset, we are interested in testing the conditions on both the GMV and tangency portfolios separately, given the limitations to unlimited lending and borrowing. We will perform these two tests on each MVE candidate portfolio (univariate KZ test with $N = 1$), as well as jointly when we test the joint hypothesis that all portfolios are spanned by the factors.

Note that all the following tables report the p -values of the F-tests while controlling for multiple testing,¹⁷ which is more restrictive than the p -values from the original test of Kan and Zhou (2012) (Appendix C.1 explains in detail the steps of the multiple testing method).

4.1. Mean–variance spanning test of the traditional multi-factor models

We assume the case of a typical investor who follows (i) the market model of Sharpe (1964), or (ii) the Fama and French (1993) 3-factor model. We test the superiority of the portfolio sort configurations for risk-optimization strategies for these two investor profiles.

In the case of the first investor, we follow the empirical framework of Kan and Zhou (2012) and start with an efficient frontier made of the market benchmark and a long-term maturity bond. This corresponds to the typical allocation of a pension fund. The CAPM represents the coordinates of a line made of these two assets in a quantity equal to β and $(1-\beta)$.¹⁸ Therefore, R_1 comprises two assets, i.e., an investment in a 30-year U.S. treasury bond (B30) and the market portfolio.¹⁹

In the case of the second investor, we consider two scenarios. First, we consider that they will expand their original portfolio (R_1) made of the 30-year U.S. treasury bond (B30) and the market portfolio (Mkt) of the 2×3 size and book-to-market basis portfolios, as obtained from Kenneth French's website, forming the size and value risk premia from the Fama and French (1993) 3-factor

¹⁶ The MATLAB code is available on Prof. Guofu Zhou's website.

¹⁷ We control for multiple testing given the large number of MVE candidates.

¹⁸ Please note that the test of spanning could not work for $K=1$, i.e., with a benchmark portfolio made of a single asset.

¹⁹ The gross return of the market portfolio (Mkt) is obtained by adding the risk-free rate (RF) to the excess returns of the market portfolio ($Mkt - RF$) based on the data from Kenneth French's website, while the 30-year U.S. treasury bond (B30) is obtained from the CRSP U.S. Treasury and Inflation Indexes.

Table 6
Spanning tests with multiple factors: Full sample.

Models	(1)			(2)			(3)			(4)		
	α	F_1	F_2	α	F_1	F_2	α	F_1	F_2	α	F_1	F_2
Panel A: Independent												
MD ₂ × 3	0.21%	12.95***	1.07	0.02%	1.00	15.41***	0.02%	0.76	22.70***	−0.11%	4.03**	10.09***
MD ₃ × 3	0.24%	13.39***	1.54	0.04%	3.15	25.41***	0.03%	2.26	29.03***	−0.12%	3.59**	11.49***
MD ₃ × 3 × 3	0.21%	7.88**	0.55	−0.03%	0.75	1.72	−0.04%	0.96	0.97	−0.06%	0.78	19.59***
MV ₂ × 3	0.37%	17.09***	8.26**	0.08%	1.71	13.84***	0.08%	2.01	38.27***	−0.13%	2.15	14.81***
MV ₃ × 3	0.33%	12.89***	4.49	0.06%	0.94	11.83**	0.06%	1.20	34.74***	−0.09%	1.02	20.33***
MV ₃ × 3 × 3	0.33%	16.95***	7.34*	0.06%	1.49	20.40***	0.07%	2.26	73.44***	−0.13%	2.82*	12.11***
RP ₂ × 3	0.22%	14.05***	0.55	0.03%	3.96	18.74***	0.03%	3.73	34.71***	−0.12%	4.61**	12.06***
RP ₃ × 3	0.23%	12.05***	0.06	0.03%	4.08	13.52***	0.03%	3.96*	26.70***	−0.11%	2.99*	18.92***
RP ₃ × 3 × 3	0.24%	12.39***	0.00	0.02%	1.08	3.20	0.02%	1.09	7.14**	−0.11%	2.96*	21.91***
Long-only MVE												
MVE ₂ × 3	0.48%	20.41***	1.65	0.31%	11.07***	2.82	0.30%	10.93***	6.06	−0.14%	2.18	36.53***
MVE ₃ × 3	0.53%	21.81***	1.59	0.34%	12.79***	3.13	0.33%	11.75***	3.01	−0.15%	2.34	38.77***
MVE ₃ × 3 × 3	0.49%	17.71***	2.25	0.35%	12.61***	6.12*	0.33%	11.76***	8.47*	−0.11%	1.15	37.89***
Panel B: Dependent												
MD ₂ × 3	0.37%	17.10***	10.93**	0.17%	13.52***	74.36***	0.15%	9.43**	71.01***	−0.13%	2.76*	9.02**
MD ₃ × 3	0.42%	15.42***	18.21***	0.23%	12.08***	84.79***	0.19%	7.92**	78.32***	−0.09%	0.94	10.97***
MD ₃ × 3 × 3	0.47%	13.80***	16.92**	0.25%	8.71***	66.03***	0.20%	5.41*	55.59***	−0.04%	0.13	20.00***
MV ₂ × 3	0.58%	21.41***	18.96***	0.32%	12.79***	55.36***	0.28%	9.75***	59.90***	−0.10%	0.79	22.11***
MV ₃ × 3	0.54%	19.51***	23.95***	0.28%	10.90***	67.14***	0.23%	7.40**	62.79***	−0.09%	0.68	16.05***
MV ₃ × 3 × 3	0.42%	14.38***	16.63***	0.20%	7.51**	54.41***	0.18%	5.87**	71.83***	−0.07%	0.45	15.31***
RP ₂ × 3	0.30%	13.30***	5.74*	0.14%	14.00***	73.14***	0.12%	10.72***	85.54***	−0.11%	1.97	11.78***
RP ₃ × 3	0.33%	11.33***	6.81**	0.16%	12.33***	81.05***	0.14%	8.37***	78.19***	−0.07%	0.63	16.73***
RP ₃ × 3 × 3	0.36%	12.93***	5.68*	0.18%	14.34***	67.72***	0.15%	10.35***	66.06***	−0.09%	1.00	18.49***
Long-only MVE												
MVE ₂ × 3	0.85%	41.84***	5.10	0.65%	36.41***	13.06***	0.61%	33.13***	14.03**	−0.26%	5.46**	42.89***
MVE ₃ × 3	0.92%	42.13***	7.34**	0.74%	42.91***	20.74***	0.70%	38.83***	23.98***	−0.24%	4.18**	44.71***
MVE ₃ × 3 × 3	0.73%	24.62***	7.18*	0.63%	31.39***	27.77***	0.59%	27.46***	30.59***	−0.11%	0.88	43.79***

The table reports the results for the bootstrap mean–variance spanning test from Kan and Zhou (2012). We test the improvement brought to the tangent (F_1) or to the GMV (F_2) portfolio when a test asset (R_2) is added to the benchmark assets (R_1). The test is performed twice in Panel A and B, given that we have two proxies for R_1 , that is, the smart beta strategies performed on the set of $i \times j$ independent- or dependent-basis portfolios. The outcomes of the test are (i) the abnormal return of the candidates (α), (ii) the F-tests, and (iii) the bootstrap p-values that control for multiple testing. The regression models are as follows:

- (1) $R_1 = MKT + B30$, $R_2 = SB_{i \times j}$
- (2) $R_1 = MKT + B30 + SL + SM + SH + BL + BM + BH$, $R_2 = SB_{i \times j}$
- (3) $R_1 = MKT + SL + SM + SH + BL + BM + BH$, $R_2 = SB_{i \times j}$
- (4) $R_1 = SB_{i \times j} + B30$, $R_2 = Mkt$

The results presented below are composed of 1000 simulations for each smart beta (SB) strategy, i.e., maximum diversification (MD), minimum variance (MV), risk parity (RP), and the long-only maximum Sharpe ratio (MVE). All the strategies are based on raw returns. B30 refers to the 30-Year U.S. Treasury Bonds. The size and book-to-market (2×3) portfolios denoted as small-low B/M (SL), small-medium B/M (SM), small-high B/M (SH), big-low B/M (BL), big-medium B/M (BM), and big-high B/M (BH) are retrieved from Kenneth French's website. The sample period is composed of monthly returns from July 1963 to December 2015.

*Denote significance levels of 10%.

**Denote significance levels of 5%.

***Denote significance levels of 1%.

model. Second, we consider the case in which investors hold an equity-only allocation with the market portfolio and the 6 size and book-to-market basis portfolios.

In all three of these scenarios, R_2 is the gross return of one smart beta strategy (i.e., raw returns that do not take transaction costs into account).²⁰

We consider a last scenario (4) wherein the investors hold R_1 , which comprises the 30-year U.S. treasury bond (B30) and the gross return of one smart beta strategy. Contrary to the other scenarios, the test asset, or R_2 , is now the market portfolio (Mkt).

The step-down spanning test proceeds as follows: we first test the null hypothesis H_0^1 that the α is equal to 0, meaning that no improvement is obtained in the efficient frontier by adding the test asset to the initial benchmark portfolio (R_1). We provide the F-statistics (F_1) and consider the usual significance thresholds, i.e., 1%, 5%, and 10%. We also use the F-test on the GMV portfolio (F_2).

²⁰ We consider the strategies' gross returns (i.e., before transaction costs) because the risk factors used as explanatory variables are computed before transaction costs. We provide further evidence for the performance of the strategies' net of transactions in the next sub-section of the paper.

We report in Table 6 the results for the dependent and independent opportunity sets. Only the bootstrapped p -values are reported, as these results control for multiple testing, consequently making them more conservative than the standard p -values found in the MVE test from Kan and Zhou (2012).

In Model (1), while the tangent portfolio level (F_1) is significantly improved when adding a smart beta strategy with a 99% confidence level for almost all strategies when using both the dependent- and independent-basis portfolios, the GMV portfolio (F_2) is rarely improved when using independent portfolios, as shown in Panel A. This means that the traditional CAPM model does not span an expanded set augmented with risk-optimization strategies on dependent-basis portfolios (Panel B), while it does for independent-basis portfolios. Additionally, we observe that all the alphas and F -statistics using the dependent-basis portfolios (Panel B) are superior to those using the independent portfolios (Panel A).

In Model (2), investors expand their opportunity set made of the market portfolio and the bond component with the 2×3 size and book-to-market basis portfolios. The results of Model (2) indicate that a smart beta strategy executed on dependent-basis portfolios shifts the investor initial efficient frontier (Panel B), which is contrary to the addition of independent smart betas. Taking the average alpha across smart beta strategies leads, on yearly basis, to a 2.62% (resp. 0.39%) additional risk-adjusted return when using dependent (resp. independent)-basis portfolios.

The same results hold for Model (3) when the bond component is not integrated into the benchmark portfolio. We further test the sensitivity of our results to the bond maturity in Appendix E.

Finally, Model (4) shows that the market portfolio provides negative alphas over each smart beta strategy performed either on an independent or a dependent basis.

To compare our results to Grinblatt and Saxena (2018)'s framework, the last three lines of each panel also provide the special case of the long-only MVE strategy. This strategy improves the tangent portfolios for both the dependent and independent frameworks. However, the superiority of the dependent framework is twofold. First, the mean–variance spanning tests are not rejected for the independent smart beta, given the non-significance of F_2 . Second, the mean–variance spanning tests show that the long-only MVE strategies performed on dependent-basis portfolios provide alphas about 2 times higher than those under the independent framework. This large outperformance is due to a better diversification but also a return improvement that cannot be achieved by using risk-based strategies (as evidenced in Table 5 of Section 3.2).

We replicate the results from Table 6 in Appendix E for the following variants: (i) we test the period after the publication date of the seminal Fama and French (1993), as suggested in Mclean and Pontiff (2016) (see Table E.1), (ii) we replace the 30-year bond with a 10-year bond (see Tables E.2 and E.3), and finally, (iii) we use a 60-month estimation window for estimating the covariance matrix (see Table E.4).

All these robustness tests support our main results. We first show that the superiority of the dependent smart beta strategies is not related to the choice of bond maturity or the testing period. Not only are our results significant, but the alphas from the independent framework are also economically lower. In variant (iii), although the outperformance of the dependent- over independent-basis portfolios in forming smart beta strategies still holds, the alphas and their significance are slightly reduced. This result is expected, as the matrix of variance–covariance is time-dependent, which means that a shorter window is necessary to provide more up-to-date estimates.

4.2. Horse race between dependent- and independent-basis portfolios

The previous subsection suggests that dependent-basis portfolios offer better properties to perform risk-based optimization. We therefore carry out a horse race between the opportunity sets made of basis portfolios formed after dependent and independent sorting. The spanning test considers whether a portfolio (R_1) composed of the U.S. government 30-year bonds and a smart beta formed on the sorting configuration A spans this set of portfolios (R_1) plus the same smart beta but performed on the sorting configuration B. In our applications, this leads to two different scenarios. In Scenario 1, R_2 is a smart beta formed on the independent-sorted portfolios (SB_{ind}), while in Scenario 2, R_2 is the same smart beta but performed on the dependent-sorted portfolios (SB_{dep}). Here, the smart betas are based on raw returns but *net* transaction costs. Details on the estimation of the transaction costs can be found in Appendix D. Both H_1^0 (the test on the tangency portfolio) and H_2^0 (the test on the GMV) are tested, and only the bootstrapped p -values are reported, as they control for testing both candidates simultaneously and, consequently, are more conservative than the standard p -values.

The first part of Table 7 performs the horse race in a univariate setting and compares the properties of the dependent- and independent-basis portfolios for each risk-based optimization.

In Model (1), we test whether the low-risk strategies formed on dependent opportunity sets span a larger universe augmented with independent sets. For all low-risk strategies, we cannot reject mean–variance spanning at the 10% confidence level. This means that the efficient frontier comprising a low-risk optimization of dependent portfolios and an investment in a long-term U.S. government bond cannot be improved using an independent opportunity set. However, Model (2) indicates that the MD (2×3 , 3×3 , and $3 \times 3 \times 3$) and MV (2×3 , 3×3) strategies performed on a dependent opportunity set improve both the tangency and the GMV portfolios formed on an independent opportunity set. This is evidenced by the levels of p -values attached to F -tests on H_1^0 and H_2^0 when the dependent portfolio is used as R_2 (right part of the table). Empirically, the best improvement is found for the $MD_{3 \times 3 \times 3}^{dep}$ with a monthly abnormal net return of 0.23% (2.75% annually) over a combination of the long-term U.S. bond and $MD_{3 \times 3 \times 3}^{ind}$. The univariate tests also show the outperformance of the long-only MVE portfolio constructed using a dependent opportunity set.

In the second part of Table 7, we use basis portfolios as benchmark assets in R_1 and group all the smart betas in R_2 . Model (3) refers to smart betas constructed on independent-basis portfolios, while Model (4) refers to smart betas constructed on dependent-basis portfolios. We use basis portfolios with the regular splits (2×3 , 3×3 , and $3 \times 3 \times 3$). The superiority of the smart betas

Table 7
Horse Race Bootstrap test.

	Model (1)			Model (2)		
	α	F_1	F_2	α	F_1	F_2
$MD_2 \times 3$	−0.05%	0.80	0.06	0.13%	5.82**	11.17***
$MD_3 \times 3$	−0.01%	0.02	0.12	0.14%	4.16**	17.58***
$MD_3 \times 3 \times 3$	0.00%	0.00	2.47	0.23%	5.96**	18.90***
$MV_2 \times 3$	0.03%	0.12	6.52**	0.20%	5.25**	10.17***
$MV_3 \times 3$	−0.01%	0.04	0.09	0.19%	6.10**	24.21***
$MV_3 \times 3 \times 3$	0.08%	1.49	3.21*	0.06%	0.79	9.72***
$RP_2 \times 3$	0.00%	0.01	0.30	0.06%	1.74	5.81**
$RP_3 \times 3$	0.01%	0.04	0.12	0.07%	1.50	9.12***
$RP_3 \times 3 \times 3$	0.00%	0.01	0.09	0.08%	2.32	9.24***
<u>Long-only MVE</u>						
$MVE_2 \times 3$	−0.13%	2.28	8.25***	0.39%	18.97***	5.88**
$MVE_3 \times 3$	−0.11%	1.37	7.50***	0.40%	17.39***	7.80***
$MVE_3 \times 3 \times 3$	0.06%	0.40	9.83***	0.27%	5.90**	9.70***
<u>Joint tests</u>						
	Model (3)			Model (4)		
	F-GRS	F_1	F_2	F-GRS	F_1	F_2
$Ind_2 \times 3$	1.74	1.95	5.20***	3.86***	3.91***	6.53***
$Ind_3 \times 3$	1.98	2.11	3.04*	3.76***	3.71***	5.33***
$Ind_3 \times 3 \times 3$	1.94	1.86	3.13**	3.35**	3.22**	6.53***
$Dep_2 \times 3$	1.96	1.80	5.61***	3.01***	2.42**	4.40***
$Dep_3 \times 3$	1.97	1.83	4.90***	2.93**	2.44**	3.37**
$Dep_3 \times 3 \times 3$	1.95	1.78	3.63***	2.73*	2.28*	4.47***
<u>Joint tests</u>						
	Model (5)			Model (6)		
	F-GRS	F_1	F_2	F-GRS	F_1	F_2
2×3	1.84	1.71	9.46***	7.65***	9.05***	9.58***
3×3	2.04*	1.98*	7.99***	6.12**	6.89***	8.36***
$3 \times 3 \times 3$	1.21	1.20	2.38**	2.75***	3.25***	3.35***

Models (1) and (2) report the results of the bootstrap mean–variance spanning test from Kan and Zhou (2012). F_1 (resp. F_2) tests the improvement in the tangent (resp. GMV) portfolio when the test asset (R_2) is added to the benchmark assets (R_1). Each smart beta (SB) strategy is the net of the transaction costs (TCs), which are estimated as in Hasbrouck (2009). We report the abnormal return (α) of the test asset candidates, the F-tests, and the bootstrap p-values over 1000 runs. The regression models are defined with a 30-Year U.S. Treasury Bonds (B30) as follows:

- (1) $R_1 = B30 + SB_{i \times j}^{dep}$ net of TCs, $R_2 = SB_{i \times j}^{ind}$ net of TCs
- (2) $R_1 = B30 + SB_{i \times j}^{ind}$ net of TCs, $R_2 = SB_{i \times j}^{dep}$ net of TCs

Models (3) to (6) report two multivariate joint tests with the F-GRS test and the KZ spanning test for a joint improvement of both the tangency (F_1) and the GMV portfolio (F_2) when all R_2 are added to the benchmark assets R_1 . The benchmark assets are composed of $i \times j$ basis portfolios obtained with a sorting (Sort) method of either dependent (Dep) or independent (Ind) splits. The F-GRS uses returns in excess of the risk-free rate, while the KZ test uses raw returns. The regression models are as follows:

- (3) $R_1 = B30 + \text{all Sort}_{i \times j}$ basis portfolios, $R_2 = \text{all } SB_{i \times j}^{ind}$ net of TCs
- (4) $R_1 = B30 + \text{all Sort}_{i \times j}$ basis portfolios, $R_2 = \text{all } SB_{i \times j}^{dep}$ net of TCs
- (5) $R_1 = B30 + \text{all Dep}_{i \times j}$ basis portfolios, $R_2 = \text{all Ind}_{i \times j}$ basis portfolios
- (6) $R_1 = B30 + \text{all Ind}_{i \times j}$ basis portfolios, $R_2 = \text{all Dep}_{i \times j}$ basis portfolios

The sample period is composed of monthly returns from July 1963 to December 2015.

*Denote significance levels of 10%.

**Denote significance levels of 5%.

***Denote significance levels of 1%.

constructed on dependent-basis portfolios is twofold: either from the joint GRS test – on the joint significance of the alphas from all smart betas when regressed on a set of basis portfolios – or from the multivariate spanning test à la KZ, which evaluates the joint improvement of both the tangency (F_1) and the GMV portfolio (F_2).

Additionally, we present direct evidence of the mean–variance superiority of the dependent-basis portfolios over the independent-basis portfolios under Models (5) and (6). For instance, the first row under the specification “ 2×3 ” of Model (5) means that R_1 is composed of a long-term bond and the 6 (2×3) size and value *dependent*-basis portfolios, while R_2 is composed of the 6 (2×3) size and value *independent*-basis portfolios. Under Model (6), R_1 refers to a long-term bond and the 6 (2×3) size and value *independent*-basis portfolios, while R_2 refers to the 6 (2×3) size and value *dependent*-basis portfolios. The results of Model (6) provide support

for significant mean–variance improvement when using dependent-basis portfolios, regardless of the number of basis portfolios, by rejecting both the GRS and KZ tests.²¹

5. MVE benchmark selection

We follow the method of [Harvey and Liu \(2021\)](#) to select the most appropriate (without luck) MVE benchmark among the low-risk portfolios and the original CW portfolio to explain the cross-section of expected returns. Our test assets are the 2×3 and 3×3 portfolios sorted based on size and the book-to-market ratio or the $3 \times 3 \times 3$ when the sorting procedure is first pre-conditioned on a firm's momentum. The MVE benchmark should best complement a basis multi-factor model comprising a long-term U.S. Government rate as a proxy for the risk-free rate (B30) and the size (SMB) and the value (HML) factors of [Fama and French \(1993\)](#).

The method is an alternative to the test developed by [Gibbons et al. \(1989\)](#), which fails to measure the relative performance of two competing models or the incremental explanatory power of one additional factor when both models generate intercepts different from 0. This is due to the fact that it assumes that the model is optimal under the null hypothesis (H0).

To measure the incremental contribution of the selected candidate, [Harvey and Liu \(2021\)](#) define a scaled intercept (SI) measure and examine the spread between the scaled (by the standard error of the estimated intercept) intercept of the augmented and initial models. Using equivalent notations to those of the authors, the measure is defined as follows:

$$SI_{ew}^{med} = \frac{\text{median}(\{|a_i^g|/s_i^b\}_{i=1}^J) - \text{median}(\{|a_i^b|/s_i^b\}_{i=1}^J)}{\text{median}(\{|a_i^b|/s_i^b\}_{i=1}^J)} \quad (12)$$

where $\text{median}(\cdot)$ is the median value of the ratio $|a_i^g|/s_i^b$ or $|a_i^b|/s_i^b$. Here the superscript b represents the baseline model, and g represents the augmented model. The subscript i refers to the i -th portfolio among the J test assets, and s denotes the standard errors of the regression intercept a .

A negative value of the SI means that the augmented model outperforms the baseline model in explaining the variations of the J test asset returns. To define a statistical level of confidence in the measure, [Harvey and Liu \(2021\)](#) used the bootstrapping method presented in Step 2 of [Appendix C.1](#).

To orthogonalize the MVE candidates, the authors regress the returns of R_2^i , where i denotes the i -th candidate among the list of K candidates, against the baseline benchmark R_1 and then subtract the intercept from the time-series R_2^i , as follows:

$$\begin{aligned} R_2^i &= \alpha^i + \beta^i R_1 + e^i \\ R_2^{\alpha,i} &= R_2^i - \alpha^i = \beta^i R_1 + e^i \end{aligned} \quad (13)$$

In our applications, R_1 is composed of the 30-Year U.S. Bond (B30), size (SMB), and value (HML) factors. $R_2^{\alpha,i}$ is defined as a linear combination of the benchmark assets (R_1), i.e., the 30-Year U.S. Bond (B30), size (SMB) and value (HML) factors, such that it does not bring any additional information to the baseline model.

Then, in each sample of the B bootstrap, a score for the scaled intercept SI_{ew}^{med} can be obtained for the K number of orthogonalized candidates (i.e., $R_2^{\alpha,i}$ with the $i = \{1, 2, \dots, K\}$ candidates). Hence, the single-test p -value for the i -th candidate is given by:

$$p\text{-val} = \frac{\#\{SI^o > SI^b\}}{B} \quad (14)$$

Our results might be subject to multiple testing since 7 MVE candidates are tested on the same data. To control for multiple testing, we follow [Harvey and Liu \(2021\)](#) and take the minimum value among K estimates of SI in the b -th bootstrap, as follows:

$$SI^{b,*} = \min_{i \in \{1, 2, \dots, K\}} \{SI^{b,i}\} \quad (15)$$

Hence, the multiple-test p -value for the i -th candidate is written as follows²²:

$$p\text{-val} = \frac{\#\{SI^o > SI^{b,*}\}}{B} \quad (16)$$

Next, our objective is to apply the method to a multiple set of MVE candidates and filter them to find the best candidate. For example, the first natural candidate to consider is the traditional cap-weighted market portfolio. However, smart beta strategies on Fama–French's independent 2×3 size and value portfolios or on the dependent 2×3 size and value portfolios can also constitute MVE candidates to augment the baseline model. We also extend the 2×3 size and value grid to a 3×3 or $3 \times 3 \times 3$ splits with an additional sort based on the firms' momentum and end up with a set of 7 candidates for the MVE portfolio. We run the test sequentially, as in [Harvey and Liu \(2021\)](#), until the single test p -value of each candidate is greater than a pre-specified threshold. In our application, we set the threshold to 10%. In each run, the selected candidate has a single-test p -value but will also be attributed a multiple-test p -value to control for data snooping. The candidate is only accepted if the multiple test p -value is significant at the 90% confidence level.

²¹ We thank the Referee for suggesting this additional test to provide further support for the importance of dependent sorting.

²² Note that the sign of the indicator function is important. Here, we want to count the number of bootstrapped scaled intercepts (SI^b) that have lower values (improvement of the model) than the scaled intercept from the original sample (SI^o). In other words, when the test is performed on the time-series of R_2 from the original data.

Table 8 presents the results for the different types of basis assets and smart beta portfolios. The table shows the single-test p -value for each MVE candidate, as well as the final joint p -value for the selected candidates considering the multiple testing framework. Note that except for the cap-weighted market portfolios, all the smart beta portfolio candidates are net transaction costs, as estimated in Appendix D.1. For the 2×3 portfolios, the optimal MVE candidate comes from the same family as the set of basis portfolios to be explained, i.e., independent for Panel A and dependent for Panel B. However, as soon as the dimensions of the sort increase and, therefore, the dispersion between portfolios increases, the dependent candidates win the horse race (Panel D to F).

In summary, our results can be explained by two elements documented by recent academic research. First, Grinblatt and Saxena (2018) show that the weighting scheme and the 2×3 framework underlying the Fama and French (1993) methodology make “rank deficient” factors that fail to price the basis assets. Our method provides an alternative weighting scheme based on an extended 3×3 or $3 \times 3 \times 3$ opportunity set, which produces factors more integrative of the basis portfolios. Second, the imbalance of the distribution of stocks in portfolios under an independent sort is too sensitive to a number of macro-economic factors (Daniel and Titman, 2012; Lambert et al., 2020) and leads to the construction of sub-optimal basis portfolios.

6. Alphabet portfolios

This section investigates the role of the underlying characteristics in building basis portfolios. Our objective is to illustrate the impact of the sorting method when the underlying characteristics command a significant relationship with future stock returns. To this end, we construct smart betas on “Alphabet” portfolios instead of portfolios sorted across size, value and momentum dimensions. These “Alphabet” portfolios are formed based on random characteristics, which are defined by the letters found in the tickers of stocks. More precisely, at the end of June each year, we assign a random value obtained from a standard normal to each letter of the alphabet and allocate stocks into basis portfolios according to these random values at the beginning of July. The first characteristic is based upon the first letter of the ticker, while the second characteristic is related to the second letter, and so on.²³ Thus, the characteristics should be time-varying in a random manner.

The underlying construction methodology for the basis portfolios follows the independent sort with NYSE breakpoints of Fama and French (1993) and the dependent sort with whole-sample breakpoints of Lambert et al. (2020). We form cap-weighted portfolios using 2×3 , 3×3 , and $3 \times 3 \times 3$ splits. Next, we allocate these basis portfolios into one final smart beta strategy (i.e., risk parity, minimum variance, and maximum diversification), which rebalances the basis portfolios every month. Finally, we run the bootstrap mean–variance spanning test from KZ described in the previous sections and verify whether these random allocations have significantly better Sharpe ratios than the CW market portfolio and the underlying basis portfolios used in the Fama and French (1993) 3-factor model. In short, we want to test whether the definition of the underlying characteristics matters when constructing smart betas on basis portfolios.

By definition, the purpose of sorting into portfolios resides in factoring characteristics into returns for which the characteristics command a linear relationship with $E[R]$. For instance, a *positive* relationship between a characteristic and the expected return would imply that $E[R_i] < E[R_j]$, where $i < j$, corresponding to the i -th and j -th portfolios. However, if the characteristic does not command any relationship with the expected return, we should obtain $E[R_i] \approx E[R_j] \approx \beta E[R_m]$ and $\sigma[R_i] \approx \sigma[R_j] \approx \sigma[R_m]$ when the number of basis portfolios increases. Consequently, any smart betas (linear combination) of the N basis portfolios should not systematically outperform the market portfolio.

However, the results displayed in Model (1) of Table 9 show that smart betas outperforming the market portfolio can still be found when basis portfolios are formed on random characteristics, i.e., when alphas are positive and F_1 are significant and even when controlling for the size and value basis portfolios underlying the Fama and French (1993) 3-factor model, as evidenced by the results in Model (2). Nonetheless, contrasting these results with Model (2) from Table 6, we can formulate three remarks for smart betas on random portfolios, as follows: the alphas are (i) not dependent on the sorting methods, (ii) are at least 2x lower than those for sorts based on determinant characteristics, and (iii) are greater for risk parity optimizations. When $\sigma[R_i] \approx \sigma[R_j]$, risk parity optimizations attribute a weight to each portfolio that is close to $1/N$, and consequently, the source of the abnormal return earned over the cap-weighted market portfolio might be attributed to the monthly rebalancing of the random portfolios rather than to the choice of the allocation itself, as shown in Plyakha et al. (2015).

In summary, the test demonstrates that sorting methods are important when stock characteristics are correlated with each other and are related to the expected return, while they are not when the characteristics are uncorrelated with each other. Additionally, smart betas constructed on random signals can only outperform the cap-weighted market portfolio by maintaining constant weights and frequent rebalancing. This evidence might thus be helpful in filtering the large universe of ETFs, among which some smart betas are potentially designed based on “fake” signals and only rely on frequent rebalancing schemes.²⁴

7. Concluding remarks

New tendencies have emerged in passive investing toward smart beta strategies and style investing as a channel by which to obtain mean–variance efficient portfolios through low-risk objectives. Our paper reconciles the trends by applying long-only risk-based strategies to characteristic-sorted equity portfolios and addresses the mean–variance efficiency of these products. The exercise

²³ If the ticker is missing in the database, we use the first letters of the company name from CRSP (COMNAM).

²⁴ Here, fake signals refer to characteristics that might correlate with the expected return in-sample but fail to do so out-of-sample or post-publication, as has been documented in the work of Mclean and Pontiff (2016), among others.

Table 8
Lucky MVE candidates.

MVE Candidates →	Mkt	MV _{dep}	MD _{dep}	RP _{dep}	MV _{ind}	MD _{ind}	RP _{ind}	Mkt	MV _{dep}	MD _{dep}	RP _{dep}	MV _{ind}	MD _{ind}	RP _{ind}
	Baseline = B30 + SMB + HML								Baseline = B30 + SMB + HML					
	Panel A: 2 × 3 cap-weighted independent portfolios as test assets								Panel B: 2 × 3 cap-weighted dependent portfolios as test assets					
GRS	4.836	4.189	4.391	4.538	4.155	4.445	4.341	12.058	10.947	11.344	11.589	11.552	11.888	11.962
p-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Scaled intercept (SI)	0.042	0.070	0.041	0.036	0.066	−0.012	−0.842	0.049	0.061	−0.838	0.023	0.009	−0.001	−0.008
Single test <i>p</i> -value	0.701	0.984	0.929	0.894	0.893	0.325	0.000	0.880	0.865	0.000	0.782	0.569	0.547	0.468
SI sequence	2	2	2	2	2	2	1	2	2	1	2	2	2	2
Selected candidate(s)	RP _{ind}							MD _{dep}						
Multiple test <i>p</i> -value	[0.000]							[0.000]						
	Panel C: 3 × 3 cap-weighted independent portfolios as test assets								Panel D: 3 × 3 cap-weighted dependent portfolios as test assets					
GRS	3.403	2.792	2.892	2.975	2.909	2.918	2.929	8.228	7.757	7.959	7.950	8.116	8.108	8.108
p-value	0.000	0.003	0.002	0.002	0.002	0.002	0.002	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Scaled intercept (SI)	−0.006	0.041	0.255	0.373	−0.876	0.100	0.169	0.210	0.249	0.221	−0.806	0.001	0.150	0.107
Single test <i>p</i> -value	0.542	0.755	0.883	0.895	0.000	0.697	0.755	0.937	0.979	0.916	0.000	0.528	0.861	0.849
SI sequence	2	2	2	2	1	2	2	2	2	2	1	2	2	2
Selected candidate(s)	MV _{ind}							RP _{dep}						
Multiple test <i>p</i> -value	[0.000]							[0.000]						
	Panel E: 3 × 3 × 3 cap-weighted independent portfolios as test assets								Panel F: 3 × 3 × 3 cap-weighted dependent portfolios as test assets					
GRS	2.262	2.135	2.164	2.213	2.085	2.332	2.110	4.591	4.366	4.335	4.283	4.335	4.608	4.379
p-value	0.000	0.001	0.001	0.000	0.001	0.000	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Scaled intercept (SI)	0.245	0.030	−0.790	0.301	0.283	0.111	0.225	−0.015	0.007	−0.668	−0.039	0.006	0.013	0.020
Single test <i>p</i> -value	0.943	0.661	0.000	0.970	0.962	0.816	0.952	0.566	0.571	0.000	0.431	0.610	0.686	0.739
SI sequence	2	2	1	2	2	2	2	2	2	1	2	2	2	2
Selected candidate(s)	MD _{dep}							MD _{dep}						
Multiple test <i>p</i> -value	[0.000]							[0.000]						

The table reports a sequential test developed by [Harvey and Liu \(2021\)](#) to select robust MVE candidates. As MVE candidates, we use the traditional cap-weighted market portfolio obtained from Kenneth French's website and the smart beta strategies constructed on the two sets of basis portfolios, as follows: one from an (ind)ependent sort and the other from a (dep)endent sort. All strategies are taken in excess of the risk-free rate and are net transactions costs. The test assets used in the MVE test are the basis portfolios used to construct the smart beta strategies, which are indicated next to the referenced panel. The baseline model used in our test is the 30-Year U.S. Treasury Bonds in excess of the risk-free rate (B30) and the size (SMB) and value (HML) factors obtained from Kenneth French's website. In the first row of each panel, we report the [Gibbons et al. \(1989\)](#) test, with its respective *p*-value listed in the second row; the third row presents the value of the scaled intercept (SI); the fourth row displays the single-test *p*-value for SI; the fifth row shows the numbering sequence to which SI and its *p*-values refer; the penultimate row reports the selected MVE candidates among the smart beta strategies; and the final row displays the *p*-values of the selected candidate when controlling for multiple testing. The sample period ranges from July 1963 to December 2015.

Table 9
Spanning tests with multiple factor: Alphabet portfolios.

Models	(1)			(2)		
	α	F_1	F_2	α	F_1	F_2
Panel A: Independent						
$MD_{2 \times 3}$	0.04%	7.15**	0.10	0.02%	1.98	0.20
$MD_{3 \times 3}$	0.04%	5.29**	0.42	0.01%	0.68	1.79
$MD_{3 \times 3 \times 3}$	0.05%	3.97*	2.67	0.02%	0.42	2.15
$MV_{2 \times 3}$	0.02%	0.20	1.41	−0.03%	0.96	0.31
$MV_{3 \times 3}$	0.07%	4.50*	0.27	0.07%	4.43**	0.24
$MV_{3 \times 3 \times 3}$	0.04%	0.74	0.10	0.03%	0.50	0.28
$RP_{2 \times 3}$	0.05%	9.33***	0.16	0.02%	2.87	0.17
$RP_{3 \times 3}$	0.05%	8.79***	0.66	0.02%	1.74	2.53
$RP_{3 \times 3 \times 3}$	0.07%	10.10***	0.96	0.03%	2.55	0.82
Panel B: Dependent						
$MD_{2 \times 3}$	0.03%	2.28	0.34	0.01%	0.07	0.80
$MD_{3 \times 3}$	0.05%	10.26***	0.19	0.03%	3.09	1.08
$MD_{3 \times 3 \times 3}$	0.06%	6.56**	8.57***	0.03%	1.65	9.70***
$MV_{2 \times 3}$	0.03%	0.55	1.26	−0.01%	0.09	2.08
$MV_{3 \times 3}$	0.05%	2.08	0.89	0.03%	0.62	1.01
$MV_{3 \times 3 \times 3}$	0.01%	0.06	3.47	−0.01%	0.05	3.84
$RP_{2 \times 3}$	0.03%	2.22	0.40	0.00%	0.05	0.95
$RP_{3 \times 3}$	0.06%	11.97***	0.01	0.03%	3.16*	0.78
$RP_{3 \times 3 \times 3}$	0.07%	8.30***	10.93***	0.03%	1.88	12.61***

The table reports the results for the bootstrap mean–variance spanning test from Kan and Zhou (2012) when basis portfolios are based on the “alphabet” characteristics. The mean–variance test is performed as follows: we test whether a benchmark portfolio R_1 has a significant improvement at the tangent (F_1) or at the GMV (F_2) portfolio level when a test asset (R_2) is added to the benchmark assets (R_1). The test is performed twice in Panel A and B, given that we have two proxies for R_1 , that is, the smart beta strategies performed on the set of $i \times j$ independent- or dependent-basis portfolios. The outcomes of the test are as follows: (i) the abnormal return of the candidates (α), (ii) the F-tests, and (iii) the bootstrap p-values that control for multiple testing. The regression models are as follows:

$$(1) \quad R_1 = MKT + B30, \quad R_2 = SB_{i \times j}$$

$$(2) \quad R_1 = MKT + B30 + SL + SM + SH + BL + BM + BH, \quad R_2 = SB_{i \times j}$$

The results presented below are composed of 1000 simulations for each smart beta (SB) strategy, i.e., maximum diversification (MD), minimum variance (MV), risk parity (RP), and the long-only maximum Sharpe ratio (MVE). All the strategies are based on raw returns. B30 refers to the 30-Year U.S. Treasury Bonds. The size and book-to-market (2×3) portfolios denoted as small-low B/M (SL), small-medium B/M (SM), small-high B/M (SH), big-low B/M (BL), big-medium B/M (BM), big-high B/M (BH) are from Kenneth French’s website. The sample period is composed of monthly returns from July 1963 to December 2015.

*Denote significance levels of 10%.

**Denote significance levels of 5%.

***Denote significance levels of 1%.

is economically important for two reasons. First, the recent inflation of the discovered risk factors questions the capitalization-weighted market portfolio as a mean–variance efficient candidate. Second, there is a common practice among institutional investors to reallocate funds across style groupings (e.g., Froot and Teo, 2008).

We show that the methodology for grouping stocks in different style buckets has substantial implications for the performance of the selected smart beta strategy. To categorize stocks into investment style portfolios, we stratify the universe along the academic standard dimensions of size, value, and momentum characteristics. We implement two sorting methodologies: (a) a dependent sort on all-breakpoints and (b) an independent sort on NYSE-breakpoints. We demonstrate through a set of state-of-the-art mean–variance tests that risk optimizations on dependent portfolios achieve a superior performance after controlling for transaction costs and multiple testing. Contrary to conventional wisdom, our results show that the trade-off between allocating more weight to smaller capitalization stocks and transaction costs can be offset by the benefit of diversification achieved by a sophisticated sorting method, i.e., a dependent sort. Because a dependent sort controls for correlated variables and stratifies the stock universe in well-diversified portfolios (Lambert et al., 2020), this sorting methodology delivers significant diversification benefits for smart beta strategies. We substantiate this point by extending the diversification return approach of Booth and Fama (1992). Economically, we infer from our findings that our method, which reconciles style investing with smart beta strategies, not only reduces the curse of dimensionality in portfolio optimizations but is also well-aligned with current practices in the mutual fund industry.

This study thus contributes to the development of novel sorting methods to obtain efficient basis portfolios, which can improve the mean–variance performance of smart beta strategies.

CRedit authorship contribution statement

Boris Fays: Conceptualization, Methodology, Software, Formal analysis, Data curation, Visualization, Writing - original draft, Writing - review & editing. **Nicolas Papageorgiou:** Conceptualization, Validation, Writing - review & editing, Supervision. **Marie**

Lambert: Conceptualization, Methodology, Validation, Formal analysis, Resources, Writing - original draft, Writing - review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This paper has benefited from comments by Vikas Agarwal, Wale Dare, Georges Hübner, Astrid Salzmann (discussant), Xiaoqing Eleanor Xu (discussant), as well as the participants in the 34th International Conference of the French Finance Association (Valence), the Financial Management Association 2018 (San Diego), the Quant Vision Summit 2018 (QMI, Paris) and the participants in research seminars at the University of Aachen, University of Paris-Dauphine, HEC Montreal, and Ghent University. We would also like to thank an anonymous referee for insightful suggestions.

Funding

Marie Lambert thanks Deloitte Belgium and Luxembourg for their financial support and acknowledges financial support from Quantitative Management Research Initiative (QMI) under the aegis of the Fondation du Risque, a joint initiative by Université Paris-Dauphine, l'école Nationale de la Statistique et de l'Administration ParisTech (ENSAE), and LFIS Capital.

Appendix A. Estimation of the covariance matrix

In this section, we briefly describe a shrinkage methodology used in our applications to estimate the covariance with lower sampling errors following [Ledoit and Wolf \(2004\)](#). In their model, the authors build on [Elton and Gruber \(1973\)](#), who use a constant correlation coefficient to shrink the assets' covariance toward a global average correlation estimator.

The constant correlation coefficient is determined as follows:

$$\hat{\rho} = \frac{1}{N(N-1)} \left(\sum_i^N \sum_j^N \hat{\rho}_{ij} - N \right) \quad (\text{A.1})$$

where N is the number of portfolios—in our applications, 6, 9 or 27. The term $\hat{\rho}_{ij}$ is the historical correlation estimate between the i -th portfolio and the j -th portfolio. [Ledoit and Wolf \(2004\)](#) then obtain an optimal structure for the covariance matrix and reduce the sampling error of a traditional sample covariance matrix (S) as follows:

$$\Sigma = \delta F + (1 - \delta)S \quad (\text{A.2})$$

where Σ is the output covariance matrix obtained from the shrinkage estimation and δ is the optimal shrinkage intensity.²⁵ S is the sample covariance matrix from our 60 daily returns, and F is the structured covariance matrix with the assets' covariance estimated via the constant-correlation estimator in Eq. (A.1).²⁶ In our empirical study, the estimations of the sample and the structured covariance matrices are based on 60-day rolling windows to accommodate gradual changes in the return distribution and short-term variations. A real-life application with tradable assets ([Idzorek and Kowara, 2013](#)) would impose constraints on the historical information available to replicate our results. For this reason – to stay as close as possible to what real-world applications may offer – we limit our optimizations with 60-day windows. This choice is also consistent with [Fama and French \(2018\)](#), who estimate the monthly variance of stocks using 60 days of lagged returns.

Appendix B. Testing the incremental diversification return

[Ledoit and Wolf \(2008\)](#) propose an “indirect” bootstrap methodology to construct an empirical distribution of the spread in a function of the underlying first and second moments of two time-series. They test the significance of the spread by considering whether a $1-\alpha$ confidence interval (e.g., 90%) contains zero.

The authors first consider that the difference between the true first and second moments of the two series converge towards their sample estimate such that

$$\sqrt{T}(\hat{u} - u) \xrightarrow{d} N(0, \Omega) \quad (\text{B.1})$$

where $\hat{u} = (\hat{\mu}_i, \hat{\mu}_j, \hat{\sigma}_i^2, \hat{\sigma}_j^2)$ are the sample estimates of $u = (\mu_i, \mu_j, \sigma_i^2, \sigma_j^2)$, $\xrightarrow{d}$ refers to the convergence in the distribution of the parameters, T is the length of the time-series, and Ω refers to the variance of the estimator distribution.

²⁵ The MATLAB code is available at Prof. Wolf's [website](#).

²⁶ The covariance of matrix F is given by $\sigma_{ij} = \hat{\rho}_{ij} \sigma_i \sigma_j$.

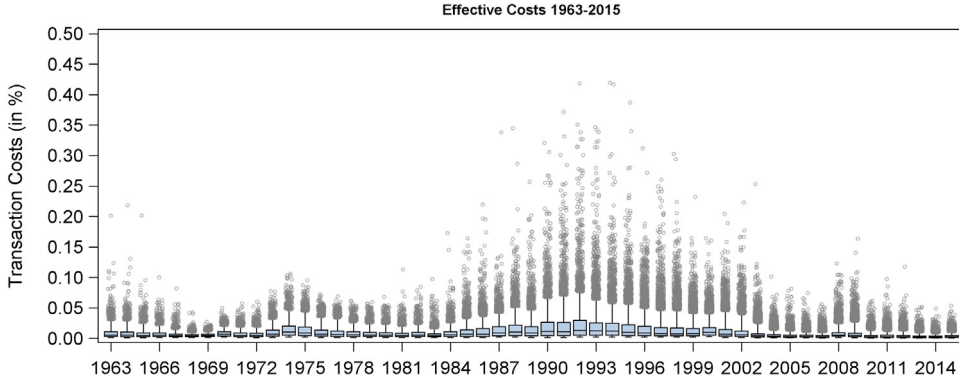


Fig. D.4. Variation in Transaction Cost Estimates. The figure presents a boxplot of the distribution of individual stock transaction costs estimated as in [Hasbrouck \(2009\)](#). The sample period is the interval from 1963 to 2015. The whiskers represent the distribution of the 5th to 95th percentile, and the upper and lower edges of the boxes correspond to the 25th and 75th percentiles. The gray dots represent outliers.

Considering the sample uncentered second moments instead of the sample estimated variances, i.e., $\hat{\gamma}_i = E(r_i^2)$ and $\hat{\gamma}_j = E(r_j^2)$, and taking into account non-normality and auto-correlation in returns, the relationship (B.1) becomes the following:

$$\sqrt{T}(\hat{v} - v) \xrightarrow{d} N(0, \Psi) \quad (\text{B.2})$$

where $\hat{v} = (\hat{\mu}_i, \hat{\mu}_j, \hat{\gamma}_i, \hat{\gamma}_j)$ is the sample estimate of $v = (\mu_i, \mu_j, \gamma_i, \gamma_j)$.

The estimator Ψ is estimated through a heteroskedasticity and autocorrelation (HAC) robust kernel method. We refer to the paper of [Ledoit and Wolf \(2008\)](#) for a more detailed discussion on the computation of this estimator.

The standard error of the spread $\hat{\Delta}$ in a function $f(\hat{v})$ can be defined as follows:

$$s(\hat{\Delta}) = \sqrt{\frac{\nabla' f(\hat{v}) \hat{\Psi} \nabla f(\hat{v})}{T}} \quad (\text{B.3})$$

where $\nabla' f(\hat{v})$ is the gradient function of $f(\hat{v})$ and T is the length of the time-series.

To obtain a confidence interval attached to $\hat{\Delta}$, we resample the original time-series using the block-bootstrap method of [Politis and Romano \(1992\)](#) and construct an empirical (bootstrap) distribution of a studentized test statistic (d^b) defined as follows:

$$d^b = \frac{|\hat{\Delta}^b - \hat{\Delta}|}{s(\hat{\Delta}^b)} \quad (\text{B.4})$$

where the superscript b denotes the b -th bootstrap sample and where $s(\hat{\Delta}^b)$ for the b -th bootstrap is obtained by using both the gradient of $f(\hat{v}^b)$ and the HAC kernel estimator $\hat{\Psi}^b$ and is defined as follows:

$$s(\hat{\Delta}^b) = \sqrt{\frac{\nabla' f(\hat{v}^b) \hat{\Psi}^b \nabla f(\hat{v}^b)}{T}} \quad (\text{B.5})$$

The bootstrap $1-\alpha$ confidence interval is defined as follows:

$$\left[\hat{\Delta} - z_{| \cdot |, 1-\alpha/2}^b s(\hat{\Delta}), \hat{\Delta} + z_{| \cdot |, 1-\alpha/2}^b s(\hat{\Delta}) \right] \quad (\text{B.6})$$

where $z_{| \cdot |, 1-\alpha}^b$ is the quantile of the distribution function of the studentized statistic estimated from the bootstrap and is denoted as $\mathcal{L}(d^b)$.

In our applications, we use a block-bootstrap of 10 observations and run 4999 simulations.²⁷ The bootstrap process works as follows: First, we set a length for the block of observations (e.g., 10) that we want to resample to capture serial autocorrelation. Second, we match the length of the original time-series in the bootstrap samples to preserve the uncertainty and degrees of freedom from the original data. Third, we randomly resample (with replacement) the sequence of the time-series for the b -th bootstrap and keep the same sequence for resampling the time-series of the strategies and their underlying opportunity sets. This way, we make sure to preserve the cross-sectional correlation across the assets (see, e.g., [Fama and French, 2010](#); [Harvey and Liu, 2021](#)). Finally, we repeat the operation B times, e.g., 4999 times, to construct an empirical distribution of centered studentized test statistics in which the standard error defining a studentized test statistic (d) of the original time-series is as follows:

$$d = \frac{|\hat{\Delta}|}{s(\hat{\Delta})} \quad (\text{B.7})$$

²⁷ Our results are not sensitive to the choice of the block length. As a matter of fact, we ran the test with blocks of lengths {2, 4, 6, 8, and 10} and found very similar results (available upon request). Additionally, we ran 4999 simulations to stay aligned with the recommendations of [Ledoit and Wolf \(2008, p. 858\)](#).

The p -value attached to the test of the spread $\hat{\Delta}$ in a function $f(\hat{v})$ is computed as follows:

$$p\text{-val} = \frac{\#\{d^b \geq d\} + 1}{B + 1} \quad (\text{B.8})$$

Ledoit and Wolf (2008) apply this framework to a test of the Sharpe ratio for a pair of strategies. They consider the following function:

$$f(a, b, c, d) = \frac{a}{\sqrt{c - a^2}} - \frac{b}{\sqrt{d - b^2}} \quad (\text{B.9})$$

where $a = \hat{\mu}_i$, $b = \hat{\mu}_j$, $c = \hat{\gamma}_i$, and $d = \hat{\gamma}_j$.

The gradient of the function is defined as $\nabla' f(\hat{v}) = \left(\frac{c}{(c-a^2)^{1.5}}, -\frac{d}{(d-b^2)^{1.5}}, -\frac{1}{2} \frac{a}{(c-a^2)^{1.5}}, \frac{1}{2} \frac{b}{(d-b^2)^{1.5}} \right)$ and is the gradient function of $f(\hat{v})$.

Our estimates of $\hat{\Delta}$ are defined as follows:

$$\begin{aligned} \hat{\Delta}(DR) &= DR^{Dep} - DR^{Ind} \\ \hat{\Delta}(DR_1) &= DR_1^{Dep} - DR_1^{Ind} \\ \hat{\Delta}(DR_2) &= DR_2^{Dep} - DR_2^{Ind} \end{aligned} \quad (\text{B.10})$$

However, obtaining the gradient for these functions is cumbersome because in addition to the pair of strategies, we are left with a number N of dependent-sorted and independent-sorted portfolios. Given that in our applications, this amount N can take the value of 6, 9 or 27, finding the gradient for this large number of parameters is difficult. Consequently, we make the assumption that if there are large deviations in the diversification return spread for a pair of strategies, then there should also be large deviations in their Sharpe ratio spread. This assumption is helpful, as we can now only substitute the numerator in Eqs. (B.4) and (B.7) based on the spread in the diversification return while keeping the standard error from the spread in the Sharpe ratio derived in the initial framework of Ledoit and Wolf (2008). To test the sensitivity of this assumption, we also substitute the standard error from the spread in the Sharpe ratio with the standard error from the spread in the geometric return, as well as with the standard error of the spread in the geometric return scaled by the standard deviation.

Their gradient functions ($\nabla' f(\hat{v})$) are respectively given by $(a + 1, -b - 1, -0.5, 0.5)$ and $\left(\frac{0.5a^3 - 0.5ac - c}{a^2 - c\sqrt{c-a^2}}, -\frac{0.5b^3 + 0.5bd + d}{b^2 - d\sqrt{d-b^2}}, \frac{-0.25a^2 + 0.5a + 0.25c}{a^2 - c\sqrt{c-a^2}}, \frac{0.25b^2 - 0.5b - 0.25d}{b^2 - d\sqrt{d-b^2}} \right)$. We obtained qualitatively similar results under all the robustness tests performed. The results are available upon request.

Appendix C. Mean-variance spanning test

The KZ step-down test proceeds as follows: The first test defines the null hypothesis for the tangent portfolio such that $\alpha = 0_N$ using the OLS regression. The tangency portfolio is improved when the null hypothesis is rejected.

$$H_0^1 : \alpha = 0_N \quad (\text{C.1})$$

Kan and Zhou (2012) perform a test for the statistical significance of the hypothesis similar to a GRS F -test. The F -test for the first hypothesis (H_0^1) is as follows:

$$F_1 = \frac{T - K - N}{N} \left[\frac{\hat{a} - \hat{a}_1}{1 + \hat{a}_1} \right] \quad (\text{C.2})$$

where T is the number of observations; K is the number of benchmark assets; N is the number of test assets; $\hat{a}_1 = \hat{\mu}_1' \hat{V}_{11}^{-1} \hat{\mu}_1$ represents the squared Sharpe ratio of the K benchmark assets (R_1), with $\hat{V}_{11}$ denoting the variance and $\hat{\mu}_1$ the vector of the mean return of the benchmark assets; and $\hat{a}$ takes the same notation as $\hat{a}$ but refers to the benchmark assets plus the new test asset (R).

The second test of the step-down procedure defines the null hypothesis for the GMV portfolio. This second test is conditional on the first test, $\alpha = 0_N$, and verifies whether $\delta = 1_N - \beta 1_K = 0_N$. Only when both conditions are rejected does the test suggest that the GMV portfolio is improved by adding N assets to the K benchmark assets.

$$H_0^2 : \delta = 1_N - \beta 1_K = 0_N | \alpha = 0_N \quad (\text{C.3})$$

The F -test for the second hypothesis (H_0^2) is as follows:

$$F_2 = \frac{T - K - N + 1}{N} \left[\frac{\hat{c} + \hat{d}}{\hat{c}_1 + \hat{d}_1} \frac{1 + \hat{a}_1}{1 + \hat{a}} - 1 \right] \quad (\text{C.4})$$

where $\hat{c}_1 = 1_K' \hat{V}_{11}^{-1} 1_K$ and $\hat{d}_1 = \hat{a}_1 \hat{c}_1 - \hat{b}_1^2$ are the efficient set (hyperbola) constants, with $\hat{a}_1 = \hat{\mu}_1' \hat{V}_{11}^{-1} \hat{\mu}_1$ and $\hat{b}_1 = \hat{\mu}_1' \hat{V}_{11}^{-1} 1_K$ for the benchmark assets (R_1). $\hat{\mu}_1$ and $\hat{V}_{11}$ denote the vector of the mean return and the variance of the benchmark assets, respectively. $\hat{a}$, $\hat{b}$, $\hat{c}$ and $\hat{d}$ are the equivalent notations for the benchmark assets plus the new test assets (R).

Mean-variance spanning implies that both null hypotheses hold (H_0^1 and H_0^2). The benchmark assets R_1 are said to span the test assets R_2 if the weight attributed to the N test assets within the efficient frontier comprising $K + N$ assets is trivial. In other words, discarding the N test assets does not significantly change the efficient frontier of the K benchmark assets from a statistical standpoint. By testing the two hypotheses separately, we gain an understanding of the reason for the mean-spanning rejection. If the mean-variance test is rejected, the test assets improve either the slope of the tangency portfolio or the risk-return properties of the GMV portfolio. Assuming the existence of a risk-free rate, investors are mostly concerned with the difference in the tangency portfolios.

C.1. Multiple test: A bootstrap approach

To test the robustness of our results, we extend the mean–variance spanning tests to address the multiple testing concern. We implement the bootstrap method used in [Harvey and Liu \(2021\)](#). The method proceeds in 4 steps, as follows:

Step 1: Orthogonalization Under the Null

The goal of this step is to modify the original time series of R_2 such that the null hypothesis appears to be true in-sample ([Harvey and Liu, 2021](#); [White, 2000](#)). To do this, we perform the following regression:

$$R_2^t = \alpha + \beta R_1^t + R_{2,e}^t \quad (\text{C.5})$$

Then, we can work on the last equation to obtain an orthogonal time-series, as denoted by the subscript $\perp$, that satisfies $Qw_1 = 0$ and $Qw_2 = 0$ while preserving the dependence between R_2 and R_1 , as follows:

$$R_{2,\perp}^t = \beta_{\perp} R_1^t + R_{2,e}^t \quad (\text{C.6})$$

where the term $\beta_{\perp} = \frac{\beta}{\beta' 1_K}$ is a simple re-scale of the original vector of the slopes (β) that satisfies $\delta = 1 - \beta_{\perp}' 1_K = 0$. Moreover, one can easily identify that the last equation also satisfies $\alpha = 0$. We use this new time-series $R_{2,\perp}$ in our bootstrap to estimate the statistical validity of the F-values from both hypothesis tests (H_0^1 and H_0^2). Note that only H_0^2 is a joint test where $\delta = 1_N - \beta 1_K = 0_N$ is conditional on $\alpha = 0_N$.

Step 2: Bootstrap

The bootstrap procedure is a random selection of monthly observations of the strategies with replacement (i.e., R_1 and $R_{2,\perp}$). We jointly resample the monthly observations to preserve the cross-sectional correlations across strategies, as in [Fama and French \(2010\)](#). Additionally, we ensure that the new time-series is the same size as the original time frame (630 months) to ensure that the degrees of freedom in the measurements of the bootstrap F-tests remain equal to the F-test from the original sample ([Harvey and Liu, 2021](#)).

Step 3: MVE spanning test

We apply the mean–variance spanning from [Kan and Zhou \(2012\)](#) to the bootstrapped samples according to the benchmark assets (R_1) and the test asset ($R_{2,\perp}$). We repeat the operation B times (1000) to construct an empirical distribution of the performance measures. In these bootstrap samples, the null is valid in-sample, and a significant value for the F-tests simply arises from the resampling (or luck). The empirical distribution serves as a threshold for the critical value of the F-tests. Each bootstrap contains four F-tests, i.e., two for the tangent ($F_{1,ind}^b$ and $F_{1,dep}^b$) and two for the GMV ($F_{2,ind}^b$ and $F_{2,dep}^b$), with the subscript b denoting the b -th bootstrapped sample, while *ind* and *dep* denote the independent and dependent-sorted portfolios.

Step 4: Controlling for multiple testing

To control for multiple testing, we follow the framework of [Harvey and Liu \(2021\)](#) and adjust the confidence intervals of the original F-tests by keeping the maximal measure of each hypothesis (tangent and GMV) for each bootstrap. For instance, for the tangent hypothesis, the reference point for the b -th bootstrap is $F_1^b = \max(F_{1,ind}^b, F_{1,dep}^b)$. Hence, we solve the multiple testing issue by comparing the distribution of the B maximal statistic measures to that found in the original sample $F_{1,ind}^o$ and $F_{1,dep}^o$, where o denotes the original sample.

The frequency of observations in the bootstrap sample that are greater than the F-test under the original sample defines the bootstrap p -value. Thus, the p -value is the sum of the indicator value $I\{F_1^o < F_1^b\}$ divided by the total number of bootstraps B .

Appendix D. Transaction costs

To consider transaction costs, we follow an approach similar to that of [Novy-Marx and Velikov \(2016\)](#) and use the individual stock estimates from the Gibbs sampling developed in [Hasbrouck \(2009\)](#). This approach is practically useful, as we trade stocks on NYSE-NASDAQ-AMEX exchanges and consequently have to differentiate between transaction costs for small- and large-cap stocks.

[Novy-Marx and Velikov \(2016\)](#) uncover a minor drawback to Hasbrouck's estimation technique, which requires relatively long series of daily prices to perform the estimation (250 days), resulting in a number of missing observations (mostly for non-NYSE stocks), for which the authors perform a non-parametric matching method and attribute equivalent transaction costs to the stock with a missing value according to its closest match to a stock with a non-missing value based on their size and idiosyncratic volatility. However, according to the authors, these missing observations represent only 4% of the total market capitalization universe. Instead, we replace the missing values with a transaction cost of 0.50%. We employ this value because (1) we see from [Fig. D.4](#) that only a very small number of estimates from Hasbrouck's algorithm have breached a trading cost of 50 bps since 1963, (2) this choice will more strongly impact illiquid stocks with a small number of daily observations (small-capitalization stocks), and (3) [Plyakha et al. \(2015\)](#) also chose to set this threshold for transaction costs from 1993 onwards.

In [Fig. D.4](#), we show the annual box-and-whisker plot for the CRSP/Gibbs estimates of transaction costs (variable c from Eq. (D.4)) from 1963 to 2015.

The next subsection describes the estimation process of the effective costs, as in [Hasbrouck \(2009\)](#).

Table E.1
Spanning tests with multiple factors: Subsample.

Models	(1)			(2)			(3)			(4)		
	α	F ₁	F ₂	α	F ₁	F ₂	α	F ₁	F ₂	α	F ₁	F ₂
Panel A: Independent												
MD ₂ × 3	0.21%	4.55*	5.58*	0.05%	1.60	11.14**	0.04%	1.32	20.59***	−0.07%	0.51	3.55**
MD ₃ × 3	0.27%	6.03**	6.14*	0.12%	7.18**	16.65***	0.10%	5.78**	24.32***	−0.09%	0.77	4.57**
MD ₃ × 3 × 3	0.18%	2.36	5.01	0.00%	0.00	4.11	−0.01%	0.05	4.33	0.01%	0.01	7.58**
MV ₂ × 3	0.45%	11.16***	17.21***	0.19%	4.27	15.61***	0.19%	4.25*	38.67***	−0.14%	0.93	5.72**
MV ₃ × 3	0.42%	8.99**	9.59***	0.21%	5.13*	10.43**	0.22%	5.82**	36.40***	−0.10%	0.44	11.43***
MV ₃ × 3 × 3	0.47%	13.66***	15.51***	0.24%	8.81***	16.94***	0.24%	9.11***	47.25***	−0.19%	2.00	4.90**
RP ₂ × 3	0.23%	5.45*	4.00	0.05%	4.85*	10.74**	0.05%	5.13*	31.40***	−0.09%	0.80	5.05**
RP ₃ × 3	0.26%	5.65**	2.72	0.08%	10.03***	8.03**	0.08%	10.25***	23.49***	−0.09%	0.67	8.44***
RP ₃ × 3 × 3	0.29%	6.98**	3.47	0.08%	7.32**	5.35*	0.08%	6.75**	10.22***	−0.11%	1.03	7.93***
Long-only MVE												
MVE ₂ × 3	0.44%	4.76**	4.18	0.26%	2.21	3.81	0.25%	2.13	8.93**	0.11%	0.37	37.33***
MVE ₃ × 3	0.42%	4.23*	4.62	0.24%	1.76	4.89*	0.20%	1.37	6.68**	0.12%	0.49	35.77***
MVE ₃ × 3 × 3	0.50%	5.15**	5.26	0.39%	4.49**	7.04*	0.36%	3.88	11.54***	0.13%	0.51	40.46***
Panel B: Dependent												
MD ₂ × 3	0.54%	15.14***	14.03***	0.45%	26.99***	36.95***	0.39%	19.53***	35.99***	−0.22%	2.42*	6.00**
MD ₃ × 3	0.55%	11.27***	14.85***	0.53%	19.63***	35.86***	0.44%	13.73***	34.46***	−0.11%	0.48	11.37**
MD ₃ × 3 × 3	0.60%	11.59***	15.05**	0.63%	19.94***	32.31***	0.53%	14.11***	30.19***	−0.09%	0.26	15.03***
MV ₂ × 3	0.79%	16.94***	15.73***	0.76%	27.60***	33.43***	0.66%	20.77***	35.26***	−0.13%	0.48	21.50***
MV ₃ × 3	0.75%	16.15***	19.40***	0.73%	24.52***	35.62***	0.62%	17.20***	31.62***	−0.13%	0.52	16.14***
MV ₃ × 3 × 3	0.57%	12.33***	17.65***	0.55%	20.33***	36.51***	0.49%	16.24***	51.58***	−0.11%	0.47	11.39***
RP ₂ × 3	0.42%	10.93***	7.70*	0.37%	29.25***	32.90***	0.33%	23.64***	43.61***	−0.16%	1.69	7.55**
RP ₃ × 3	0.43%	7.90**	7.04**	0.39%	20.61***	30.41***	0.34%	15.32***	32.16***	−0.08%	0.31	13.45***
RP ₃ × 3 × 3	0.48%	10.25***	8.23**	0.44%	26.96***	33.20***	0.38%	20.00***	33.40***	−0.13%	0.81	11.82***
Long-only MVE												
MVE ₂ × 3	0.90%	17.61***	1.43	0.81%	19.56***	2.14	0.78%	19.10***	5.72*	−0.14%	0.61	52.87***
MVE ₃ × 3	0.92%	14.84***	3.70	0.92%	21.50***	7.37**	0.88%	20.45***	15.99***	−0.02%	0.02	56.09***
MVE ₃ × 3 × 3	0.87%	11.87***	0.74	0.97%	22.45***	3.10	0.93%	21.13***	5.80**	−0.01%	0.00	66.03***

The table reports the results for the bootstrap mean–variance spanning test from [Kan and Zhou \(2012\)](#). We test the improvement brought to the tangent (F_1) or GMV (F_2) portfolio when a test asset (R_2) is added to the benchmark assets (R_1). The test is performed twice in Panels A and B, given that we have two proxies for R_1 , that is, the smart beta strategies performed on the set of $i \times j$ independent- or dependent-basis portfolios. The outcomes of the test are as follows: (i) the abnormal return of the candidates (α), (ii) the F-tests, and (iii) the bootstrap p-values that control for multiple testing. The regression models are as follows:

- (1) $R_1 = MKT + B30$, $R_2 = SB_{i \times j}$
- (2) $R_1 = MKT + B30 + SL + SM + SH + BL + BM + BH$, $R_2 = SB_{i \times j}$
- (3) $R_1 = MKT + SL + SM + SH + BL + BM + BH$, $R_2 = SB_{i \times j}$
- (4) $R_1 = SB_{i \times j} + B30$, $R_2 = Mkt$

The results presented below are composed of 1000 simulations for each smart beta (SB) strategy, i.e., maximum diversification (MD), minimum variance (MV), risk parity (RP), and the long-only maximum Sharpe ratio (MVE). All the strategies are based on raw returns. B30 refers to the 30-Year U.S. Treasury Bonds. The size and book-to-market (2×3) portfolios denoted as small-low B/M (SL), small-medium B/M (SM), small-high B/M (SH), big-low B/M (BL), big-medium B/M (BM), big-high B/M (BH) are from Kenneth French's website. The sample period is composed of monthly returns from July 1993 to December 2015.

*Denote significance levels of 10%.

**Denote significance levels of 5%.

***Denote significance levels of 1%.

D.1. Transaction costs: Gibbs estimates

A traditional model used to estimate the trading costs of a security is documented by [Roll \(1984\)](#) and simply uses the autocovariance of the change in trade price (Δp_t) to find an effective estimate of the spread such that

$$c^{Roll} = \begin{cases} \sqrt{-Cov(\Delta p_t, \Delta p_{t-1})} & \text{if } Cov(\Delta p_t, \Delta p_{t-1}) < 0 \\ 0 & \text{if } Cov(\Delta p_t, \Delta p_{t-1}) \geq 0 \end{cases} \quad (D.1)$$

In the last equation, we observe that when the autocovariance is positive, the model fails to provide a fair estimate of the effective costs. For this reason, [Hasbrouck \(2009\)](#) extends the measure under [Roll \(1984\)](#)'s framework on the price dynamics in a market with transaction costs. In this framework, the model only requires information about the daily trade price, the prior midpoint of the bid–ask prices, and the sign of the trade to perform the estimation. Formally, the price dynamic is written as follows:

$$\begin{aligned} m_t &= m_{(t-1)} + u_t \\ p_t &= m_t + c q_t \end{aligned} \quad (D.2)$$

Table E.2

Spanning tests with multiple factors: Full sample, 10-year bond.

Models	(1)			(2)			(3)			(4)		
	α	F_1	F_2	α	F_1	F_2	α	F_1	F_2	α	F_1	F_2
Panel A: Independent												
MD _{2 × 3}	0.22%	13.99***	1.62	0.02%	0.95	8.85***	0.02%	0.76	22.70***	−0.13%	5.26**	1.61
MD _{3 × 3}	0.25%	14.51***	2.20	0.04%	2.91	13.60***	0.03%	2.26	29.03***	−0.14%	4.85**	1.73
MD _{3 × 3 × 3}	0.22%	8.44***	0.83	−0.03%	0.77	1.08	−0.04%	0.96	0.97	−0.08%	1.34	4.79*
MV _{2 × 3}	0.37%	17.31***	4.53	0.07%	1.60	6.82**	0.08%	2.01	38.27***	−0.15%	2.96*	3.75*
MV _{3 × 3}	0.34%	13.89***	4.00	0.06%	1.00	7.44**	0.06%	1.20	34.74***	−0.12%	1.86	4.06*
MV _{3 × 3 × 3}	0.34%	17.06***	3.80	0.05%	1.18	7.75**	0.07%	2.26	73.44***	−0.15%	3.63**	3.19
RP _{2 × 3}	0.23%	15.20***	1.14	0.03%	3.76	9.70***	0.03%	3.73	34.71***	−0.14%	5.95***	2.17
RP _{3 × 3}	0.24%	13.32***	0.65	0.03%	3.80*	6.25**	0.03%	3.96*	26.70***	−0.13%	4.26**	3.78*
RP _{3 × 3 × 3}	0.25%	13.21***	0.17	0.02%	0.69	0.21	0.02%	1.09	7.14**	−0.13%	3.99**	5.77**
Long-only MVE												
MVE _{2 × 3}	0.51%	21.91***	2.05	0.32%	11.39***	1.91	0.30%	10.93***	6.06	−0.18%	3.52*	8.78**
MVE _{3 × 3}	0.55%	23.74***	2.51	0.35%	13.36***	2.63	0.33%	11.75***	3.01	−0.19%	3.88**	8.51***
MVE _{3 × 3 × 3}	0.52%	19.53***	3.37	0.35%	13.00***	4.25	0.33%	11.76***	8.47*	−0.15%	2.32	7.76**
Panel B: Dependent												
MD _{2 × 3}	0.39%	19.18***	11.82***	0.18%	13.59***	47.52***	0.15%	9.43**	71.01***	−0.17%	4.41**	0.23
MD _{3 × 3}	0.45%	17.74***	18.97***	0.23%	12.53***	57.74***	0.19%	7.92**	78.32***	−0.14%	2.18	0.18
MD _{3 × 3 × 3}	0.52%	16.61***	20.26***	0.27%	9.78***	52.08***	0.20%	5.41*	55.59***	−0.10%	0.92	0.89
MV _{2 × 3}	0.62%	23.81***	17.57***	0.33%	13.49***	39.11***	0.28%	9.75***	59.90***	−0.16%	1.96	2.24
MV _{3 × 3}	0.58%	21.67***	21.36***	0.29%	11.38***	46.46***	0.23%	7.40**	62.79***	−0.15%	1.74	1.02
MV _{3 × 3 × 3}	0.44%	15.85***	14.41***	0.20%	7.53**	33.63***	0.18%	5.87**	71.83***	−0.11%	1.22	1.40
RP _{2 × 3}	0.33%	15.53***	8.54**	0.15%	14.37***	48.02***	0.12%	10.72***	85.54***	−0.14%	3.55*	0.45
RP _{3 × 3}	0.37%	13.68***	10.70***	0.17%	13.00***	56.77***	0.14%	8.37***	78.19***	−0.12%	1.78	0.79
RP _{3 × 3 × 3}	0.39%	15.09***	8.56**	0.18%	14.53***	43.99***	0.15%	10.35***	66.06***	−0.13%	2.23	1.44
Long-only MVE												
MVE _{2 × 3}	0.90%	45.72***	6.53**	0.67%	38.08***	10.35***	0.61%	33.13***	14.03**	−0.33%	8.32***	7.96**
MVE _{3 × 3}	0.98%	46.83***	9.60***	0.76%	45.26***	16.71***	0.70%	38.83***	23.98***	−0.32%	7.06***	7.28**
MVE _{3 × 3 × 3}	0.81%	30.06***	14.22***	0.67%	35.03***	27.34***	0.59%	27.46***	30.59***	−0.20%	2.89*	4.27*

The table reports the results for the bootstrap mean-variance spanning test from Kan and Zhou (2012). We test the improvement brought to the tangent (F_1) or to the GMV (F_2) portfolio when a test asset (R_2) is added to the benchmark assets (R_1). The test is performed twice in Panels A and B, given that we have two proxies for R_1 , that is, the smart beta strategies performed on the set of $i \times j$ independent- or dependent-basis portfolios. The outcomes of the test are as follows: (i) the abnormal return of the candidates (α), (ii) the F-tests, and (iii) the bootstrap p-values that control for multiple testing. The regression models are as follows:

- (1) $R_1 = MKT + B10$, $R_2 = SB_{i \times j}$
- (2) $R_1 = MKT + B10 + SL + SM + SH + BL + BM + BH$, $R_2 = SB_{i \times j}$
- (3) $R_1 = MKT + SL + SM + SH + BL + BM + BH$, $R_2 = SB_{i \times j}$
- (4) $R_1 = SB_{i \times j} + B10$, $R_2 = Mkt$

The results presented below are composed of 1000 simulations for each smart beta (SB) strategy, i.e., maximum diversification (MD), minimum variance (MV), risk parity (RP), and the long-only maximum Sharpe ratio (MVE). All the strategies are based on raw returns. B10 refers to the 10-Year U.S. Treasury Bonds. The size and book-to-market (2×3) portfolios denoted as small-low B/M (SL), small-medium B/M (SM), small-high B/M (SH), big-low B/M (BL), big-medium B/M (BM), big-high B/M (BH) are from Kenneth French's website. The sample period is composed of monthly returns from July 1963 to December 2015.

*Denote significance levels of 10%.

**Denote significance levels of 5%.

***Denote significance levels of 1%.

where m_t is the log midpoint of the prior bid-ask price (the efficient price), p_t is the log trade price (the real price), q_t is the sign of the last trade of the day (+ 1 for a buy and −1 for a sale), c is the effective cost, and u_t is assumed to be unrelated to the sign of the trade (q_t).

Since we use the logarithm for the price variables in Eq. (D.2), the daily change in price is given by the following:

$$\begin{aligned} \Delta p_t &= p_t - p_{t-1} \\ &= m_t + cq_t - m_{t-1} - cq_{t-1} \\ &= c\Delta q_t + u_t \end{aligned} \quad (D.3)$$

Hasbrouck (2009) extends Roll (1984)'s model with a market factor to capture a larger part of the changes in prices not due to transaction costs. They estimate the effective trading costs using Bayesian Gibbs sampling applied to the daily prices of U.S. equities retrieved from CRSP data.²⁸ The market-factor model is presented as follows:

$$\Delta p_t = c\Delta q_t + \beta_{rm}rm_t + u_t \quad (D.4)$$

²⁸ The SAS code is available on Prof. Hasbrouck's website.

Table E.3

Spanning tests with multiple factors: Subsample, 10-year bond.

Models	(1)			(2)			(3)			(4)		
	α	F ₁	F ₂	α	F ₁	F ₂	α	F ₁	F ₂	α	F ₁	F ₂
Panel A: Independent												
MD _{2 × 3}	0.23%	5.37*	5.57*	0.05%	1.45	4.70	0.04%	1.32	20.59***	−0.09%	0.76	0.44
MD _{3 × 3}	0.29%	6.89**	5.81*	0.11%	6.26**	5.67*	0.10%	5.78**	24.32***	−0.11%	1.04	0.77
MD _{3 × 3 × 3}	0.19%	2.55	3.66	−0.01%	0.04	1.09	−0.01%	0.05	4.33	0.01%	0.00	2.60
MV _{2 × 3}	0.46%	11.59***	8.63**	0.19%	4.03	6.02*	0.19%	4.25*	38.67***	−0.15%	1.00	2.82
MV _{3 × 3}	0.45%	9.92***	6.26*	0.21%	4.99*	3.53	0.22%	5.82**	36.40***	−0.11%	0.60	4.35*
MV _{3 × 3 × 3}	0.48%	13.65***	6.78**	0.23%	7.71**	3.90	0.24%	9.11***	47.25***	−0.20%	1.95	2.95*
RP _{2 × 3}	0.26%	6.30**	4.20	0.05%	4.21	2.44	0.05%	5.13*	31.40***	−0.10%	1.09	0.94
RP _{3 × 3}	0.29%	6.63**	3.55	0.08%	8.92**	1.27	0.08%	10.25***	23.49***	−0.11%	0.98	1.80
RP _{3 × 3 × 3}	0.31%	7.53**	2.98	0.07%	5.47*	0.06	0.08%	6.75**	10.22***	−0.12%	1.20	2.49*
Long-only MVE												
MVE _{2 × 3}	0.48%	5.56**	4.06	0.27%	2.25	1.88	0.25%	2.13	8.93**	0.09%	0.25	13.46***
MVE _{3 × 3}	0.47%	5.02**	4.83	0.24%	1.78	2.83	0.20%	1.37	6.68**	0.10%	0.33	12.18**
MVE _{3 × 3 × 3}	0.55%	6.09**	5.47	0.40%	4.43*	3.47	0.36%	3.88	11.54***	0.11%	0.34	13.77***
Panel B: Dependent												
MD _{2 × 3}	0.58%	16.99***	11.84***	0.45%	25.39***	18.45***	0.39%	19.53***	35.99***	−0.25%	3.02*	1.03
MD _{3 × 3}	0.59%	12.74***	12.28***	0.52%	18.19***	17.51***	0.44%	13.73***	34.46***	−0.14%	0.72	2.84
MD _{3 × 3 × 3}	0.65%	13.27***	12.91***	0.62%	18.98***	17.06***	0.53%	14.11***	30.19***	−0.12%	0.47	3.92*
MV _{2 × 3}	0.85%	19.24***	13.07***	0.76%	27.00***	18.43***	0.66%	20.77***	35.26***	−0.17%	0.78	6.51**
MV _{3 × 3}	0.80%	17.87***	14.94***	0.73%	23.10***	18.40***	0.62%	17.20***	31.62***	−0.16%	0.75	4.79*
MV _{3 × 3 × 3}	0.60%	13.10***	10.95***	0.53%	18.28***	13.95***	0.49%	16.24***	51.58***	−0.13%	0.57	4.32*
RP _{2 × 3}	0.47%	13.03***	8.82**	0.37%	28.39***	16.31***	0.33%	23.64***	43.61***	−0.20%	2.42*	0.99
RP _{3 × 3}	0.48%	9.61***	8.63**	0.39%	19.54***	15.13***	0.34%	15.32***	32.16***	−0.11%	0.63	2.52
RP _{3 × 3 × 3}	0.52%	11.88***	8.59**	0.43%	24.83***	14.88***	0.38%	20.00***	33.40***	−0.16%	1.20	2.50
Long-only MVE												
MVE _{2 × 3}	0.96%	19.21***	1.60	0.81%	19.44***	0.66	0.78%	19.10***	5.72*	−0.17%	0.79	20.44***
MVE _{3 × 3}	0.99%	16.74***	3.90	0.93%	21.49***	3.17	0.88%	20.45***	15.99***	−0.05%	0.07	20.37***
MVE _{3 × 3 × 3}	0.98%	14.80***	3.29	1.00%	23.67***	2.39	0.93%	21.13***	5.80**	−0.06%	0.09	18.93***

The table reports the results for the bootstrap mean–variance spanning test from Kan and Zhou (2012). We test the improvement brought to the tangent (F_1) or to the GMV (F_2) portfolio when a test asset (R_2) is added to the benchmark assets (R_1). The test is performed twice in Panels A and B, given that we have two proxies for R_1 , that is, the smart beta strategies performed on the set of $i \times j$ independent- or dependent-basis portfolios. The outcomes of the test are as follows: (i) the abnormal return of the candidates (α), (ii) the F-tests, and (iii) the bootstrap p-values that control for multiple testing. The regression models are as follows:

- (1) $R_1 = MKT + B10$, $R_2 = SB_{i \times j}$
- (2) $R_1 = MKT + B10 + SL + SM + SH + BL + BM + BH$, $R_2 = SB_{i \times j}$
- (3) $R_1 = MKT + SL + SM + SH + BL + BM + BH$, $R_2 = SB_{i \times j}$
- (4) $R_1 = SB_{i \times j} + B10$, $R_2 = Mkt$

The results presented below are composed of 1000 simulations for each smart beta (SB) strategy, i.e., maximum diversification (MD), minimum variance (MV), risk parity (RP), and the long-only maximum Sharpe ratio (MVE). All the strategies are based on raw returns. B10 refers to the 10-Year U.S. Treasury Bonds. The size and book-to-market (2×3) portfolios denoted as small-low B/M (SL), small-medium B/M (SM), small-high B/M (SH), big-low B/M (BL), big-medium B/M (BM), big-high B/M (BH) are from Kenneth French's website. The sample period is composed of monthly returns from July 1993 to December 2015.

*Denote significance levels of 10%.

**Denote significance levels of 5%.

***Denote significance levels of 1%.

where rm_t is the market return on day t and β_{rm} is the parameter estimate obtained from a Bayesian regression on the market return.

The Bayesian methodology estimates the effective costs (c) based on a sequence of iterations where the initial prior for c is strictly positive and follows a normal distribution with a mean of 0.01 and a variance equal to 0.01^2 , as denoted by N^+ ($\mu = 0.01$, $\sigma^2 = 0.01^2$). This initial prior of β_{rm} follows a normal distribution with a mean and variance of 1, i.e., N ($\mu = 1$, $\sigma^2 = 1$), and the prior of σ_u^2 follows an inverted Gamma distribution initiated at IG ($\alpha = 10^{-12}$, $\beta = 10^{-12}$).²⁹ The objective of Gibbs sampling is to estimate the values of the parameters c and β_{rm} conditional upon the values drawn for q_t , which is based on the sign of the trade (Δp_t), and the error term (u_t). Initially, q_1 is set to +1 and σ_u^2 is set to 0.001. Next, the sampler runs as follows:

for 1 to 1000 sweeps

1. Perform a Bayesian OLS regression on a 250-day of lagged observations to estimate the new values of c and β_{rm} , update the posterior distribution of the parameters and make a new draw of the coefficients.

²⁹ These initial values of the priors are those found in the SAS code made available by Prof. Hasbrouck. According to Hasbrouck (2009), the initial values of the prior should not impact the final estimate of the effective cost of a stock because the first 200 iterations (of 1000) are disregarded to compute the average of the estimated values for the trading cost (c).

Table E.4

Spanning tests with multiple factors: Full sample with covariance matrix estimated on 60-Month returns.

Models	(1)			(2)			(3)			(4)		
	α	F_1	F_2	α	F_1	F_2	α	F_1	F_2	α	F_1	F_2
Panel A: Independent												
MD _{2 × 3}	0.17%	11.10***	1.22	0.03%	3.43	8.88**	0.03%	2.93	11.39***	−0.11%	4.56**	5.19**
MD _{3 × 3}	0.18%	10.07***	1.40	0.04%	2.23	6.66**	0.04%	1.93	9.36**	−0.10%	3.39*	6.63**
MD _{3 × 3 × 3}	0.19%	10.53***	0.00	0.03%	1.20	0.47	0.04%	1.31	0.31	−0.11%	3.47*	15.09***
MV _{2 × 3}	0.21%	10.89***	15.60***	0.09%	3.54*	7.19**	0.10%	4.05*	23.35***	−0.11%	2.49	0.87
MV _{3 × 3}	0.23%	12.04***	18.17***	0.11%	4.98*	10.63**	0.12%	5.37**	28.83***	−0.12%	2.88*	0.54
MV _{3 × 3 × 3}	0.22%	10.70***	8.92***	0.00%	0.00	3.10	0.02%	0.08	16.38**	−0.11%	2.20	4.03**
RP _{2 × 3}	0.19%	11.91***	0.88	0.02%	4.71*	14.33***	0.02%	4.20*	21.89***	−0.11%	4.79**	6.78**
RP _{3 × 3}	0.19%	10.00***	0.13	0.02%	3.55	6.00**	0.02%	3.38	10.72***	−0.10%	3.17*	12.41***
RP _{3 × 3 × 3}	0.19%	8.86***	0.09	0.00%	0.01	1.02	0.00%	0.00	3.01	−0.09%	2.31	19.42***
Long-only MVE												
MVE _{2 × 3}	0.32%	13.51***	1.09	0.06%	1.03	1.79	0.07%	1.16	1.90	−0.13%	2.59	32.30***
MVE _{3 × 3}	0.34%	14.27***	2.09	0.09%	1.76	0.01	0.07%	1.08	3.12	−0.14%	2.77*	23.12***
MVE _{3 × 3 × 3}	0.30%	11.45***	3.09	0.08%	1.37	0.72	0.08%	1.35	1.52	−0.10%	1.56	20.79***
Panel B: Dependent												
MD _{2 × 3}	0.27%	16.27***	7.11**	0.14%	17.89***	54.05***	0.12%	14.46***	63.16***	−0.15%	5.69**	3.35
MD _{3 × 3}	0.30%	16.15***	12.55***	0.17%	15.68***	60.47***	0.15%	11.13***	53.00***	−0.16%	4.66**	2.32
MD _{3 × 3 × 3}	0.31%	13.64***	9.21**	0.15%	7.90***	38.17***	0.12%	5.03**	26.76***	−0.13%	2.93*	5.62**
MV _{2 × 3}	0.23%	14.45***	15.79***	0.10%	4.80**	10.63***	0.11%	5.48**	33.79***	−0.14%	4.63**	0.40
MV _{3 × 3}	0.24%	15.24***	16.90***	0.11%	5.74**	12.48***	0.11%	6.13**	32.79***	−0.15%	5.06**	0.25
MV _{3 × 3 × 3}	0.25%	12.11***	22.48***	0.08%	1.85	27.90***	0.07%	1.41	40.89***	−0.12%	2.49	0.40
RP _{2 × 3}	0.21%	9.51***	4.55	0.09%	9.97***	61.48***	0.08%	7.45**	71.88***	−0.10%	2.18	5.72**
RP _{3 × 3}	0.24%	7.98**	4.97*	0.11%	9.77***	71.59***	0.09%	6.24**	64.82***	−0.07%	0.85	9.76***
RP _{3 × 3 × 3}	0.25%	7.69***	2.17	0.11%	9.55***	47.66***	0.09%	6.22**	36.74***	−0.06%	0.67	15.91***
Long-only MVE												
MVE _{2 × 3}	0.40%	14.59***	0.59	0.16%	4.37*	1.31	0.16%	4.30*	2.70	−0.12%	1.70	33.51***
MVE _{3 × 3}	0.46%	16.21***	0.63	0.25%	8.11**	2.13	0.23%	6.97**	0.75	−0.13%	1.87	36.31***
MVE _{3 × 3 × 3}	0.42%	12.66***	0.62	0.30%	9.67***	4.16	0.30%	9.82***	10.66**	−0.08%	0.63	41.38***

The table reports the results for the bootstrap mean–variance spanning test from Kan and Zhou (2012). We test the improvement brought to the tangent (F_1) or to the GMV (F_2) portfolio when a test asset (R_2) is added to the benchmark assets (R_1). The test is performed twice in Panels A and B, given that we have two proxies for R_1 , that is, the smart beta strategies performed on the set of $i \times j$ independent- or dependent-basis portfolios. The outcomes of the test are as follows: (i) the abnormal return of the candidates (α), (ii) the F-tests, and (iii) the bootstrap p-values that control for multiple testing. The regression models are as follows:

- (1) $R_1 = MKT + B30$, $R_2 = SB_{i \times j}$
- (2) $R_1 = MKT + B30 + SL + SM + SH + BL + BM + BH$, $R_2 = SB_{i \times j}$
- (3) $R_1 = MKT + SL + SM + SH + BL + BM + BH$, $R_2 = SB_{i \times j}$
- (4) $R_1 = SB_{i \times j} + B30$, $R_2 = Mkt$

The results presented below are composed of 1000 simulations for each smart beta (SB) strategy, i.e., maximum diversification (MD), minimum variance (MV), risk parity (RP), and the long-only maximum Sharpe ratio (MVE). All the strategies are based on raw returns. B30 refers to the 30-Year U.S. Treasury Bonds. The size and book-to-market (2×3) portfolios denoted as small-low B/M (SL), small-medium B/M (SM), small-high B/M (SH), big-low B/M (BL), big-medium B/M (BM), big-high B/M (BH) are from Kenneth French's website. The sample period is composed of monthly returns from July 1963 to December 2015.

*Denote significance levels of 10%.

**Denote significance levels of 5%.

***Denote significance levels of 1%.

2. Back out u_t from the values of c , β_{rm} , Δp_t , rm_t , Δq_t as follows:

$$u_t = \Delta p_t - \beta_{rm} rm_t - c \Delta q_t \quad (D.5)$$

3. Update the posterior σ_u^2 according to the series of u_t ,

4. Draw a new series of q_t , knowing the new value of σ_u^2 . Given that $u_t = \Delta p_t - \beta_{rm} rm_t - c q_t + c q_{t-1}$, estimate u_t if $q_t = +1$ or $q_t = -1$. Find the probability of $u_t(q_t = +1)$ and $u_t(q_t = -1)$ given that $u_t \sim N(0, \sigma_u^2)$ and compute the odds ratio for a buy order as follows:

$$\text{Odds} = \frac{f(u_t(q_t = +1))}{f(u_t(q_t = -1))} \begin{cases} q_t = +1 & \text{if Odds} > 1 \\ q_t = -1 & \text{if Odds} < 1 \end{cases} \quad (D.6)$$

end

The process is repeated 1000 times, and the final value of c is the average of the last 800 estimations of the procedure (the “burn in” is the 200 first observations). For more information on simulating the probability distributions of q_t and u_t , as well as on the iterative process, interested readers should refer to Hasbrouck (2009, p. 1449–1951).³⁰

Appendix E. Robustness tables

Table E.1 presents the results of the mean–variance spanning tests over the sub-sample (from July 1993 to December 2015) when the proxy for the bond benchmark is the U.S. bond with a 30-year maturity.

Table E.2 presents the results of the mean–variance spanning tests over the full sample (from July 1963 to December 2015) when the proxy for the bond benchmark is the U.S. bond with a 10-year maturity.

Table E.3 presents the results of the mean–variance spanning tests over the sub-sample (from July 1993 to December 2015) when the proxy for the bond benchmark is the U.S. bond with a 10-year maturity.

Table E.4 presents the results of the mean–variance spanning tests over the full-sample (from July 1963 to December 2015) when the proxy for the bond benchmark is the U.S. bond with a 30-year maturity. Moreover, the allocation of the risk-optimization is estimated based on monthly returns with a window of 60 months.

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³⁰ Further details regarding the application of the estimation technique can also be found in Marshall et al. (2011) and Novy-Marx and Velikov (2016).

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